Linear and Discrete Optimization Modelling

UNSW MATH3171 25T3

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Contents

Chapter 1

Introduction to Optimization

1.1 Optimization: An Introduction

Many problems we counter in a large number of area entail trying to achieve the "best" outcome subject to some "restrictions". The mathematical field that encompasses these problems is called Optimization.

An optimization problem consist of three parts: decision variables, object functions and constraints.

(1) **Decision variables**: what can you change or what you wish to determine.

Denote the decision variable $\mathbf{x} \in H$ where H is a finite dimensional space. For example,

- $H = \mathbb{R}^n$ and the decision variable is a vector $\mathbf{x} \in \mathbb{R}^n$, the number of variables n.
  - Column vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$;
  - Univariate: $n = 1$;
  - Multivariate: $n = 2, 3, 4, \ldots$ up to millions;
- $H = \mathbb{R}^{p\times n}$ where $\mathbb{R}^{p\times n}$ is the set of $(p \times n)$ matrices, and the decision variable $\mathbf{x} = X \in \mathbb{R}^{p\times n}$ is an $(p \times n)$ matrix.

(2) **Objective functions**: a mathematical function of the variables quantifying the idea of "best".

- $H = \mathbb{R}^n$, the decision variable is a vector $\mathbf{x} \in \mathbb{R}^n$ and objective function $f : \mathbb{R}^n \to \mathbb{R}$. For example,
  - Linear: $f(\mathbf{x}) = \mathbf{c}^T\mathbf{x}, \quad \mathbf{c} \in \mathbb{R}^n$ fixed;
  - Affine: $f(\mathbf{x}) = \mathbf{c}^T\mathbf{x} + \gamma, \quad \mathbf{c} \in \mathbf{x}^n, \gamma \in \mathbb{R}$ fixed;
  - Quadratic: $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T G\mathbf{x} + \mathbf{c}^T\mathbf{x} + \gamma \quad G \in \mathbb{R}^{n\times n}, \mathbf{c} \in \mathbb{R}^n, \gamma \in \mathbb{R}$ fixed.
- $H = \mathbb{R}^{p\times n}$, the decision variable is an $(p \times n)$ matrix X and objective function $f : \mathbb{R}^{p\times n} \to \mathbb{R}$. For example,
  - $f(X) = \sum_{i=1}^{p}\sum_{j=1}^{n} A_{ij}X_{ij}$, where A is given $(p \times n)$ matrix X;
  - $f(X) = \text{rank}(X)$, where $\text{rank}(X)$ is the rank of a matrix X.

(3) **Constraints**: describe restrictions on the allowable values of variables.

- $H = \mathbb{R}^n$,
  - Equality constraints $g_i(\mathbf{x}) = 0, \quad i = 1, \ldots, m_E$ (e.g. binary constraint: $x_i \in \{0, 1\}$);
  - Inequality constraints $g_i(\mathbf{x}) \leq 0, \quad i = m_E + 1, \ldots, m$;
  - Geometric constraints $\mathbf{x} \in C \subseteq \mathbb{R}^n$;
  - Feasible region $\Omega \in \mathbb{R}^n$

$$\Omega = \begin{Bmatrix} \mathbf{x} \in C: & g_i(\mathbf{x}) = 0, i = 1, \ldots, m_E; \\ & g_i(\mathbf{x}) \leq 0, i = m_E + 1, \ldots, m \end{Bmatrix}$$

- The case for $H = \mathbb{R}^{p \times n}$ is similar, except the constraint functions g_i are functions on $\mathbb{R}^{p \times n}$ and the set C in the geometric constraints is a set in $\mathbb{R}^{p \times n}$. In this case, the feasible region $\Omega \subset \mathbb{R}^{p \times n}$

$$\Omega = \begin{Bmatrix} X \in C \subseteq \mathbb{R}^{p \times n}: & g_i(X) = 0, i = 1, \ldots, m_E; \\ & g_i(X) \leq 0, i = m_E + 1, \ldots, m \end{Bmatrix}$$

Optimization Problem A typical optimization problem looks like

$$\begin{aligned} \min_{x \in H} \quad & f(\mathbf{x}) \\ \text{such that} \quad & g_i(\mathbf{x}) = 0, i = 1, \ldots, m_E, \\ & g_i(\mathbf{x}) \leq 0, i = m_E + 1, \ldots, m, \\ & x \in C \subseteq H \end{aligned}$$

where H is the finite dimensional space.

Linear Programming A linear program is an optimization problem where the underlying space $H = \mathbb{R}^n$, all the functions $f, g_i, i = 1, \ldots, m$ are affine functions on $\mathbb{R}^n$ and there is no geometric constraints:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \mathbf{c}^T \mathbf{x} + \gamma \\ \text{such that} \quad & \mathbf{a}_i^T \mathbf{x} + b_i = 0, i = 1, \ldots, m_E, \\ & \mathbf{a}_i^T \mathbf{x} + b_i \leq 0, i = m_E + 1, \ldots, m. \end{aligned}$$

Conic Linear Program A conic linear program is an optimization problem where the underlying space H is a general finite dimensional space, all the functions $f, g, i = 1, \ldots, m$, are affine functions on H and the set in the geometric constraint is a specific set called convex cone.

Linear Integer Programming A linear integer programming problem is an optimization problem where the underlying space $H = \mathbb{R}^n$, all functions $f, g_i, i = 1, \ldots, m$ are affine functions on $\mathbb{R}^n$ and the geometric constraint C is contained in the set of integers:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(\mathbf{x}) \\ \text{such that} \quad & g_i(\mathbf{x}) = 0, i = 1, \ldots, m_E, \\ & g_i(\mathbf{x}) \leq 0, i = m_E + 1, \ldots, m, \\ & x \in C \subseteq \mathbb{Z}^n \end{aligned}$$

Chapter 2

Mathematical Background

2.1 Vectors and Matrices

These notes will skip some preliminaries that you should know from first and second year.

If $\mathbf{x}^T y = 0$, then we say the two vectors are $\mathbf{x}$ and $\mathbf{y}$ are orthogonal.

Definition. (Vector Norm) A *vector norm* on $\mathbb{R}^n$ is a function $\|\cdot\|$ from $\mathbb{R}^n$ to $\mathbb{R}$ such that

1. $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$ and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = 0$.
2. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. (Triangle inequality)
3. $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$ for all $\alpha \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n$.

An $(n \times n)$ matrix A is said to be symmetric if $A = A^T$. We consider eigenvalues in non-ascending order.

Theorem. (Spectral Decomposition Theorem) Let $A \in \mathbb{R}^{n \times n}$ be a symmetric $n \times n$ matrix. Then, there exists an orthogonal matrix Q (that is, $Q^TQ = QQ^T = I_n$) such that $A = QDQ^T$ where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $Q = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}$ where $\mathbf{v}_i$ is an eigenvector of A corresponding to the eigenvalue λ_i.

For a symmetric $(n \times n)$ matrix A and any $\mathbf{x} \neq 0$, we define the so-called Rayleigh quotient as

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\|\mathbf{x}\|_2^2}.$$

Then, we have, for any $\mathbf{x} \neq \mathbf{0}$,

$$\lambda_{\min}(A) \leq R_A(\mathbf{x}) \leq \lambda_{\max}(A).$$

Theorem. (Variational Characterization for Extremal Eigenvalues) For a symmetric $(n \times n)$ matrix A,

$$\lambda_{\max}(A) = \max\{R_A(\mathbf{x}) : \mathbf{x} \neq 0\} = \max_{\mathbf{x} \in \mathbb{R}^n}\{\mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\|_2 = 1\}$$

and

$$\lambda_{\min}(A) = \min\{R_A(\mathbf{x}) : \mathbf{x} \neq 0\} = \min_{\mathbf{x} \in \mathbb{R}^n}\{\mathbf{x}^T A \mathbf{x} : \|fx\|_2 = 1\}.$$

Definition. (Positive Definite Matrices) A matrix $A \in \mathbb{R}^{n\times n}$ is

- *positive definite* $\iff$ $\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0$
- *positive semi-definite* $\iff$ $\mathbf{x}^T A\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
- *negative definite* $\iff$ $\mathbf{x}^T A\mathbf{x} \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
- *indefinite* $\iff$ there exists $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n : \mathbf{x}^T A\mathbf{x} > 0$ and $\mathbf{y}^T A\mathbf{y} < 0$

A symmetric matrix $A \in \mathbb{R}^{n\times n}$

- has n real eigenvalues $\lambda_i, i = 1, \dots, n$
- Determinant: $\det(A) = \prod_{i=1}^n \lambda_i$
- Trace: $\mathrm{tr}(A) := \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$

Proposition. (Matrix Definitness) A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is

- *positive definite* $\iff$ $\lambda_i > 0$ for all $i = 1, \dots, n$
- *positive semi-definite* $\iff$ $\lambda_i \geq 0$ for all $i = 1, \dots, n$
- *negative definite* $\iff$ $\lambda_i < 0$ for all $i = 1, \dots, n$
- *negative semi-definite* $\iff$ $\lambda_i \leq 0$ for all $i = 1, \dots, n$
- *indefinite* $\iff$ there exists $i, j : \lambda_i > 0$ and $\lambda_j < 0$

Theorem. (Checking Positive (Semi-)definitness) For any eigenvalues λ of a symmetric matrix $A \in \mathbb{R}^{n\times n}$, there exists $i_0 \in \{1, \dots, n\}$ such that

$$\lambda - A_{i_0 i_0} \leq \sum_{j\neq i_0} |A_{i_0} j.$$

Definition. (Diagonally Dominant) A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is called diagonally dominant if

$$|A_{ii}| \geq \sum_{j\neq i} |A_{ij}|, \text{ for each } i \in \{1, \dots, n\}.$$

In addition, if

$$|A_{ii}| > \sum_{j\neq i} |A_{ij}|, \text{ for each } i \in \{1, \dots, n\},$$

then we say A is strictly diagonally dominant.

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix. Suppose that A is diagonally dominant and all of its diagonal elements are non-negative, then any eigenvalue of A must be non-negative, and so, A is positive semi-definite.

Definition. (Matrix Norm) A *matrix norm* on $\mathbb{R}^{p\times n}$ is a function $\|\cdot\|$ from $\mathbb{R}^{p\times n}$ to $\mathbb{R}$ such that

1. $\|X\| \geq 0$ for all $X \in \mathbb{R}^{p\times n}$ and $\|X\| = 0 \iff X = 0_{p\times n}$.
2. $\|X+Y\| \leq \|X\| + \|Y\|$ for all $X, Y \in \mathbb{R}^{p\times n}$. (Triangle inequality)
3. $\|\alpha X\| = |\alpha|\|X\|$ for all $\alpha \in \mathbb{R}, X \in \mathbb{R}^{p\times n}$.

One of the most widely used matrix norm is: **Frobenius norm**: $\|X\|_F = \left(\sum_{i=1}^{p}\sum_{j=1}^{n} X_{ij}^2\right)^{\frac{1}{2}}$.

We can also define a dot product between two matrices $A, B \in \mathbb{R}^{p\times n}$ as

$$A \cdot B = \mathrm{tr}(A^T B).$$

2.2 Local and Global Minima

Let H be a finite dimensional space and let Ω be a set in H. Let $f : H \to \mathbb{R}$ be a function on H and let $\|\cdot\|$ be a norm in H.

Definition. (Global Minimum) A point $\mathbf{x}^* \in \Omega$ is a *global minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$. The global minimum is $f(\mathbf{x}^*)$.

Definition. (Strict Global Minimum) A point $\mathbf{x}^* \in \Omega$ is a *strict global minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) < f(\mathbf{x})$ for all $\mathbf{x} \in \Omega, \mathbf{x} \neq \mathbf{x}^*$.

Definition. (Local Minimum) A point $\mathbf{x}^* \in \Omega$ is a *local minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if there exists a $\delta > 0$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$ with $\|\mathbf{x} - \mathbf{x}^*\| \leq \delta$. Then $f(\mathbf{x}^*)$ is a local minimum.

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Definition. (Closed Sets) A set Ω is said to be closed if it contains all the limits of convergent sequences of points in Ω.

Definition. (Bounded Sets) A set Ω is said to be bounded if there exists $R > 0$ such that $\Omega \subseteq \{x : \|x\| \leq R\}$.

Proposition. (Existence of Global Solution: Type I) Let H be a finite dimensional space and let Ω be a closed and bounded set in H and let f be continuous on Ω. Then the global minima of $f$over Ω exist.

Proposition. (Existence of Global Solution: Type II) Let H be a finite dimensional space and let $\|\cdot\|$ be a norm in H. Let Ω be a closed set in H and let f be continuous on Ω. Suppose that f is coercive in the sense that

$$\lim_{\|x\|\to\infty} f(x) = +\infty.$$

Then the global minima of f over Ω exist.

Relaxation Let H be a finite dimensional space. If $f : H \to \mathbb{R}$ and $\overline{\Omega} \subseteq \Omega$ then

$$\min_{x\in\Omega} f(\mathbf{x}) \leq \min_{\mathbf{x}\in\overline{\Omega}} f(\mathbf{x}).$$

2.3 Convexity for Functions/Sets in Euclidean Spaces

2.3.1 Convex Functions

Definition. A set $C \subseteq \mathbb{R}^n$ is convex if for any $\mathbf{x}, \mathbf{y} \in C$ and any $\lambda \in [0, 1]$ the following is true

$$\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in C.$$

In other words, C is convex means the line segment joining the two points $\mathbf{x} \in C$ and $\mathbf{y} \in C$ is completely contained in the set C.

Basic properties of convexity:

- Intersection of convex sets is still convex: Let $\Omega_1, \Omega_2 \subseteq \mathbb{R}^n$ be convex. Then the intersection $\Omega = \Omega_1 \cap \Omega_2$ is convex.
- Intersection of any finite number of convex sets is convex.
- Union of convex sets may or may not be convex.

Definition. (Convex Functions) Let $\Omega \subseteq \mathbb{R}^n$be a convex set. The function $f : \Omega \to \mathbb{R}$ is said to be convex if for all $\theta \in [0, 1]$ and for all $\mathbf{x}, \mathbf{y} \in \Omega$,

$$f(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta)f(\mathbf{y}).$$

f is strictly convex if and only if $f(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) < \theta f(\mathbf{x}) + (1 - \theta)f(\mathbf{y})$.

Definition. (Graident) Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable. The graident $\boldsymbol{\nabla} f : \mathbb{R}^n \to$

$\mathbb{R}^n$ of f at $\mathbf{x}$ is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n}. \end{bmatrix}$$

Definition. (Hessian) Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. The Hessian $\nabla^2 : \mathbb{R}^n \to \mathbb{R}^{n\times n}$ of f at $\mathbf{x}$ is

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix}$$

Proposition. Let $\Omega \subseteq \mathbb{R}^n$ be a convex set and $f \in C^2(\Omega)$ (f is twice differentiable on Ω).

- $\nabla^2 f(\mathbf{x})$ positive semi-definite for all $\mathbf{x} \in \Omega$ if and only if f is convex on Ω.
- If $\nabla^2 f(\mathbf{x})$ positive definite for all $\mathbf{x} \in \Omega$, then $f$is strictly convex on Ω.

Operations that preseve convexity of functions: Let Ω be a convex set.

1. Nonnegative scaling: If $f$is a convex function on Ω and $\alpha \geq 0$, then αf is also convex on Ω;
2. Sum: If f_1, f_2 are convex on Ω, then $f_1 + f_2$ is also convex on Ω;
3. Pointwise maximum: If $f_i, i = 1, \ldots, m$ are convex function on Ω, then $\max_{1\leq u\leq m} f_i$ is also convex on Ω.

Proposition. Let I be an index set and let $c_i : \mathbb{R}^n \to \mathbb{R}$ be convex functions, $i \in I$. Then

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : c_i(\mathbf{x}) \leq 0, i \in I\}$$

is a convex set.

Definition. (Convexity in the Space of Matrices) A set $C \subseteq \mathbb{R}^{p\times n}$ is convex if for any $X, Y \in C$ and any $\lambda \in [0, 1]$the following is true

$$\lambda X + (1 - \lambda)Y \in C.$$

Definition. (Convex Functions in Matirx Variable) Let $\Omega \subseteq \mathbb{R}^{p\times n}$ be a convex set. The function $f : \mathbb{R}^{p\times n} \to \mathbb{R}$ is said to be convex if for all $\theta \in [0, 1]$, and for all $X, Y \in \Omega$,

$$f(\theta X + (1 - \theta)Y) \leq \theta f(X) + (1 - \theta)f(Y).$$

Definition. Let H be a finite dimensional space. Let $f : H \to \mathbb{R}$ be a function on H and let Ω be a set in H. A **convex programming problem** has

- a convex feasible region Ω
- a convex objection function $f(\mathbf{x})$

Proposition. For a convex programming problem, any local minimizer of f on Ω is a global minimizer of f on Ω.

Chapter 3

Linear Programs

The standard form of a linear programming problem can be formulated as follows

$$\begin{aligned} \text{Minimise}_{\mathbf{x}\in\mathbb{R}^n} \quad & \mathbf{c}^T\mathbf{x} + \gamma \\ \text{subject to} \quad & \mathbf{a}_i^T\mathbf{x} = b_i, i = 1, \dots, m_E, \\ & \mathbf{x} \geq 0, \end{aligned}$$

where $\mathbf{c}, \mathbf{a}_i \in \mathbb{R}^n, b_i \geq 0, i = 1, \dots, m$ and $\gamma \in \mathbb{R}^n$. Throughout this chapter, we always assume that the feasible region is not empty.

3.1 Geometry of Linear Programming Problems

The geometry of the feasible region:

- A **hyperplane** in $\mathbb{R}^n$ is the set of all points $\mathbf{x}$ that satisfy a linear equation

  $$\mathbf{a}^T\mathbf{x} = b,$$

  for some $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$;

- A (closed) **halfspace** in $\mathbb{R}^n$ is the set of all points $\mathbf{x}$ that satisfy the linear inequality

  $$\mathbf{a}^T\mathbf{x} \leq b,$$

  for some $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$.

- The intersection of a finite number of halfspaces is called a **polyhedron**.

- A polyhedron that is bounded is called a **polytope**.

Note: (1) Finite intersection of polyhedrons is still a polyhedron. (2) Polyhedron is a convex set and the feasible region of a linear programming problem is a polyhedron. In particular, the feasible region of a linear programming problem is a convex set.

(1) The sum $\lambda_1\mathbf{x}_1 + \cdots + \lambda_k\mathbf{x}_k$ is called a **convex** combination of $\mathbf{x}_1, \dots, \mathbf{x}_k$ if the λ_i are all nonnegative and $\lambda_1 + \cdots + \lambda_k = 1$.

(2) The **convex hull** of a set S is denoted by $\operatorname{co} S$ and is defined by

$$\operatorname{co}(S) = \left\{ \lambda_1 \mathbf{x}_1 + \cdots + \lambda_k \mathbf{x}_k : \mathbf{x}_1, \ldots, \mathbf{x}_k \in S, \lambda_i \in [0,1], \sum_{i=1}^{k} \lambda_i = 1 \right\}.$$

That is, convex hull of S is the set of all convex combinations of elements of S.

Let P be a convex set in $\mathbb{R}^n$. A vector $\mathbf{x} \in P$ is a **extreme point** of P if whenever $\mathbf{x} = \lambda \mathbf{y} + (1-\lambda)\mathbf{z}$ for some points $\mathbf{y}, \mathbf{z} \in P$ and $\lambda \in [0,1]$ then $\mathbf{y} = \mathbf{z} = \mathbf{x}$.
In other words, the vector $\mathbf{x}$ cannot be written as non-trivial convex combination of two different vectors $\mathbf{y}$ and $\mathbf{z}$. Roughly speaking, $\mathbf{x}$ is an extreme point of P if $\mathbf{x}$ does not lie in any open line segment joining two points of P.

Theorem. Let P be a bounded polyhedron (or P is a polytope) in $\mathbb{R}^n$. Then, P has only finitely many extreme points, and P is a convex hull of its extreme points.

Theorem. Consider a linear programming problem P in standard form. Let Ω be the feasible region of P, that is,

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m \text{ and } \mathbf{x} \geq 0\}.$$

Suppose that Ω is bounded. Then, there must exist an optimal solution of P which is an extreme point of the feasible region Ω.

Theorem. Let Ω be the feasible region of a linear programming problem in standard formulation. Suppose that $m \leq n$ and the collection of vectors $\{\mathbf{a}_1, \ldots, \mathbf{a}_m\}$ are linearly independent. Denote $A = \begin{pmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{pmatrix} \in \mathbb{R}^{m \times n}$ and let $A_j \in \mathbb{R}^m, j = 1, \ldots, n$ be the jth column of A. Then $\mathbf{x}^* = (x_1^*, \ldots, x_n^*) \in \mathbb{R}^n$ is an extreme point of Ω if and only if the following two conditions hold

(1) $\mathbf{a}_i^T \mathbf{x}^8 = b_i, i = 1, \ldots, m, \mathbf{x}^* \geq 0$;

(2) there exists indices $1 \leq j_1, \ldots, j_m \leq n$ such that the column vector $\{A_{j_1}, \ldots, A_{j_m}\}$ is linearly independent and $x_j^* = 0$ for all $j \notin \{j_1, \ldots, j_m\}$.

3.2 Simplex Method

Definition. Consider a linear programming problem (LP) in standard form and denote its feasible region by Ω. A point $\mathbf{x}^*$ is called a **basic feasible solution** of (LP) if $\mathbf{x}^*$ is an extreme point of the feasible region Ω.

Definition. Let $\mathbf{x}$ be a basic feasible solution of the linear programming problem in standard form. The variables with zero values are called **nonbasic variables** associated with the point $\mathbf{x}$. The remaining variables are called **basic variables**.

<u>Summary of the basic simplex method</u>

1. Form the initial simplex tableau and find out an initial basic feasible solution. If this is feasible then proceed to the next step.

2. Determine the largest positive entry associated with the nonbasic variables in the cost row. This will indicate the entering basic variable. If no entries in the firsr row are positive the STOP, this is the optimal solution.

3. Determine the outgoing basic variable by finding the minimum ratio of the b entry over the (positive) entry in the column of the incoming variable. The entry in this row and column is the pivot. Use Gaussian elimination to make all other entries in this pivot column zero and divide through the pivot row by the pivot so that the pivot changes to 1. Go back to item 2.

We now introduce the two phase simplex method. In this method, we first find an initial basic feasible solution in phase 1. Then, we perform the basic simplex method to find the desired optimal solution.

1. Convert the problem to the correct form yb adding slack, surplus and artificial variables.

2. Consider an artificial problem by minimizing the sum of all artificial variables.

3. Eliminate the entries in the first row of the tableau corresponding to the basic artificial variables.

4. Use the basic simplex method on this problem. This will give an initial basic feasible solution (if it exists) for phase 2. If the solution in phase 1 has any of the artificial variables being nonzero then a feasible solution does not exist for this problem.

5. Create the initial tableau for phase 2 by using the final tableau in phase 1 and
   (a) removing all columns associated with artificial variables
   (b) replacing the cost row with the correct one for this new problem

6. Change the entries in the cost row corresponding to the basic variables to zero using the usual updating mechanism and then continue with the basic simplex method.

3.3 Duality Theory and Sensitivity Analysis

Consider the linear programming problem in standard form:

$$\begin{array}{rll} (P) & \text{Minimise}_{\mathbf{x}\in\mathbb{R}^n} & \mathbf{c}^T\mathbf{x}+\gamma \\ & \text{subject to} & \mathbf{a}_i^T\mathbf{x}=b_i, i=1,\ldots,m_E, \\ & & \mathbf{x}\geq 0, \end{array}$$

Let us call it the primal problem. The dual problem will be defined as

$$\begin{array}{rll} (D) & \text{Maximize}_{\mathbf{y}\in\mathbb{R}^m} & \gamma+\sum_{i=1}^{m} y_i b_i \\ & \text{subject to} & \mathbf{c}-\sum_{i=1}^{m} y_i\mathbf{a}_i \geq \mathbf{0} \\ & & \mathbf{y} \text{ unrestricted} \end{array}$$

Theorem. (Weak Duality) Let $\mathbf{x}$ be feasible for (P) and $\mathbf{y}$ be feasible for (D), then

$$\mathbf{c}^T\mathbf{x} \geq \mathbf{b}^T\mathbf{y} = \sum_{i=1}^{m} y_i b_i.$$

In particular, the objective function of the dual problem is a lower bound of the objective function for the primal problem.

Theorem. (Strong Duality) The following statements hold:

- If either problem (P) or problem (D) has an optimal solution then so does the other and their objective values are equal.
- If either problem has an unbounded objective value then the other problem has no feasible solution.

Theorem. (Complementary Slackness Condition) Let $\mathbf{x}^*$ be optimal for (P) and $\mathbf{y}^*$ optimal for (D), then the following conditions hold

$$x_j^*(\mathbf{c} - \sum_{i=1}^{m} y_i^* \mathbf{a}_i)_j = 0, \quad j = 1, \ldots, n.$$

Theorem. (Optimality Conditions) Let $\mathbf{x}^*$ be feasible for a primal linear programming problem (P) and let $\mathbf{y}^*$ be feasible for its dual problem (D). Then, $\mathbf{x}^*$ and $\mathbf{y}^*$ are optimal solutions for (P) and (D) respectively if and only if $(\mathbf{x}, \mathbf{y}) = (\mathbf{x}^*, \mathbf{y}^*)$ solves the following equations

$$\begin{aligned} \mathbf{a}_i^T \mathbf{x} &= b_i, i = 1, \ldots, m, \quad x \geq 0 \\ \mathbf{c} - &\sum_{i=1}^{m} y_i \mathbf{a}_i \geq 0 \\ \mathbf{c}^T \mathbf{x} &= \sum_{i=1}^{m} y_i b_i. \end{aligned}$$

3.4 Further Development of Linear Programs

3.4.1 Interior Point Methods

The basic idea of this method, is moving towards the optimum via points in the (relative)-interior of the feasible region of the linear program.

Standing Assumptions: We suppose that the feasible regions

$$F_P = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, \mathbf{x} \geq 0\}$$

and

$$F_D = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{c} - \sum_{i=1}^{m} y_i \mathbf{a}_i \geq 0\}$$

both satisfy the (relative) interior point condition, in the sense that

$$\begin{aligned} F_P^+ &= \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, \mathbf{x} > 0\} \neq \emptyset \text{ and} \\ F_D^+ &= \{\mathbf{y} \in \mathbb{R}^m : \mathbf{c} - \sum_{i=1}^{m} y_i \mathbf{a}_i > 0\} \neq \emptyset, \end{aligned}$$

where (P) and (D) are the primal and dual linear programs in standard form. We also assume that $\{\mathbf{a}_1, \ldots, \mathbf{a}_m\}$ is linearly independent.

We then construct the μ-perturbed optimality conditions, $\mu > 0$, as follows

$$\begin{cases} \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, & \mathbf{x} \geq 0 \\ \mathbf{w} = \mathbf{c} - \sum_{i=1}^m y_i \mathbf{a}_i, & \mathbf{w} \geq 0 \\ w_j x_j = \mu, & j = 1, \ldots, n. \end{cases}$$

For each $\mu > 0$, let $(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{w}(\mu))$ be a solution of the above system. Then, let

$$(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{w}(\mu)) \to (\mathbf{x}, \mathbf{y}, \mathbf{w})$$

as $\mu \to 0$. Then $(\mathbf{x}, \mathbf{y}, \mathbf{w})$ is a solution for the system above and so $\mathbf{x}$ and $\mathbf{y}$ are solutions for (P) and (D) respectively.

The sets $\{\mathbf{x}(\mu) : \mu > 0\}$ and $\{\mathbf{y}(\mu) : \mu > 0\}$ are called the **central path** of (P) and (D) respectively, where the parameter $\mu > 0$ is called the **central path parameter**.

General Interior Point Method (Outline)

- Start with an initial point $(x_0, y_0) \in F_P^+ \times F_D^+$. Let $\mu > 0$ and $\gamma \in (0, 1)$.
- Solve the following perturbed optimality system

$$\begin{cases} \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, & \mathbf{x} \geq 0 \\ \mathbf{w} = \mathbf{c} - \sum_{i=1}^m y_i \mathbf{a}_i, & \mathbf{w} \geq 0 \\ w_j x_j = \mu, & j = 1, \ldots, n. \end{cases}$$

- Replace μ by $\gamma\mu$.

3.4.2 Lagrange Multiplier Method and Necessary Optimality Conditions

Consider an optimization problem with equality constraints

$$(P_E)\text{Minimize}_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$
$$\text{Subject to} g_i(\mathbf{x}) = 0, i = 1, \ldots, m.$$

The Lagrangian function for (P_E) is

$$L(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + \sum_{i=1}^m y_i g_i(\mathbf{x}).$$

We say the Lagrange multiplier necessary optimality conditions for problem (P_E) holds at $\mathbf{x}$ if

(1) $f(\mathbf{x}) < +\infty$;

(2) $g_i(\mathbf{x}) = 0, i = 1, \ldots, m$;

(3) there exist multipliers $y_i \in \mathbb{R}, i = 1, \ldots, m$ such that

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \mathbf{y}) = \nabla f(\mathbf{x}) + \sum_{i=1}^m y_i \nabla g_i(\mathbf{x}) = \mathbf{0}.$$

We say a point $\mathbf{x}$ is a regular point for (P_E) if $\{\nabla g_i(\mathbf{x}) : 1 \leq i \leq m\}$ is linearly independent.

Proposition. (Lagrange Multiplier Necessary Optimality Conditions) For problem (P_E), let $\mathbf{x}^*$ be a local minimizer which is a regular point. Suppose that $f, g_i, i = 1, \ldots, m$ are continuously differentiable functions around $\mathbf{x}^*$. Then Lagrange multipler optimality conditions for problem (P_E) holds at $\mathbf{x}^*$, that is,

(1) $f(\mathbf{x}^*) < +\infty$;

(2) $g_i(\mathbf{x}^*) = 0, i = 1, \ldots, m$;

(3) there exist multipliers $y_i^* \in \mathbb{R}, i = 1, \ldots, m$ such that

$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \mathbf{y}^*) = \nabla f(\mathbf{x}^*) + \sum_{i=1}^{m} y_i^* \nabla g_i(\mathbf{x}^*) = \mathbf{0}.$$

3.5 Applications of Linear Programs

- Transportation problems, involving the transport of goods between various locations.
- Zero sum game theory, in particular, theory of two player zero sum games where each participant's gains or losses are exactly balanced by those of the other participants.
  - Rock-Paper-Scissors
  -

    Theorem. For any 2 person zero sum game, let P be the payoff matrix for this game. Then, we have
    $$V_1^* = V_2^* = V^* \quad \text{(called the minimax value of the game).}$$
    where
    $$V_1^* = \max_{\mathbf{x} \in S_n} \min_{\mathbf{y} \in S_m} \mathbf{y}^T P \mathbf{x}$$
    $$V_2^* = \min_{\mathbf{y} \in S_m} max_{\mathbf{x} \in S_n} \mathbf{y}^T P \mathbf{x}$$
- Radiotherapy treatment planning problem.

Chapter 4

Network Problems

Network representations of problems appear in many areas. There are electrical networks, communications networks, road networks, networks that represent project planning, networks that represent relationships between objects and quite a few more.

> **Definition. (Network)** A network contains two types of object: *nodes* and *arcs* which join the nodes. The nodes are represented by circles and the arcs by lines (in an undirected network) or arrows (in a directed network). For each arc, there is a number associated with it which is called the length of the arc. Arcs that start and finish at the same node (loops) are not allowed.

Numerically, we can recognise a directed (or undirected) network G with n nodes (labelled as $1, 2, \ldots, n$) via an associated $(n \times n)$ matrix $A = (A_{ij})_{1\leq i\leq n, 1\leq j\leq n}$ formed by

$$A_{ij} = \begin{cases} w_{ij}, & \text{if there is an arc from node } i \text{ to node } j \text{ and } w_{ij} \text{ is the corresponding arc length;} \\ 0, & \text{if there is no arc from node } i \text{ to node } j. \end{cases}$$

provided that $w_{ij} \neq 0$ for all i, j.

4.1 The Shortest Path Problem

> **Definition. (Paths and Cycles - Undirected Network Case)**
>
> - A **path** in a undirected network is a sequence of connected, distinct arcs. The length of the path is the sum of the length of the arcs traversed.
> - A **cycle** in a undirected network is a path where the starting node and end node are the same.

> **Definition. (Paths and Cycles - Directed Network Case)**
>
> - A **path** in a directed network is a sequence of connected, distinct arcs where the arcs must be traversed in the direction of the arrows, so that the tip of one arrow must join the tail of the next arrow. The length of the path is the sum of the length of the arcs traversed.
> - A **directed cycle** is a directed path where the starting node and ending node are the same.

Let the length of the arc from node i to node j be denoted by a_{ij}. If there is no arc connect from node i

to node j, let $a_{ij} = \infty$.

Dijkstra's Method (directed network)

1. Initialisation: Set $u_1 = 0$. This is the permanent label for node 1. For each node j that has an arc connecting it to node 1 set $u_j = a_{1j}$. For all other nodes set $u_j = \infty$. There are tentative labels and these nodes are assigned to the set T of tentatively labelled nodes. The set of permanent labels is $P = \{1\}$. Set iteration number $i = 0$.

2. Update the label: Find $k \in T$ where $u_k = \min_{j \in T} u_j$. Set $T = T \setminus \{k\}$ and $P = P \cup \{k\}$. If $k = n$ stop, the shortest path from node 1 to node n has been found with length u_n.

3. Update the cost: Set $u_j = \min\{u_j, u_k + a_{kj}\}$ for all $j \in T$. Update iteration number i as $i + 1$. GO back to the previous step (that is, the step of updating the label).

For an undirected network, we convert it as a directed network and apply the previous Dijkstra algorithm. In implementing the Dijkstra algorithm for undirected network, we proceeds at the same line as for directed network. The only difference is to ignore the direction of the arcs between the two nodes.

4.2 Network Flow Problem

Another important problem for networks is the network flow problem. For this kind of problem we wish to determine a collection of paths from a given starting point (called the source) to a given destination node (called the sink) whereby the amount of material sent from the starting point to the destination is a maximum.

General formula for a maximum flow problem with a single source and single sink. Denote the set of all nodes by $\mathcal{N}$ and the set of all the arcs by $\mathcal{A}$.

Definition. (Flow; Capacity; Source; Sink)

- We denote the **flow** in an arc $(i, j) \in \mathcal{A}$ by x_{ij} and this represents the amount of material going from node i to node j along that arc.

- The **capacity** c_{ij} of an arc $(i, j) \in \mathcal{A}$, where node $i \in \mathcal{N}$ is at the tail of the arc and node $j \in \mathcal{N}$ is at the tip of the arc, is the maximum amount of goods that can be sent through that arc in that direction. We will assume that the minimum amount that can be sent through is zero.

- A **source** is a node where the flow out of that node exceeds the flow into the node. A source is often denoted by s.

- A **sink** is a node where the flow into that node exceeds the flow out of the node. A sink is often denoted by t.

- The arc (i, j) is said to be *outgoing* from node i, *incoming* to node j, and *incident* to both i and j. Define

$$I(i) = \{j \in \mathcal{N} : (j, i) \in \mathcal{A}\} \text{ and } O(i) = \{j \in \mathcal{N} : (i, j) \in \mathcal{A}\}.$$

Conservation law of the network: For each node i where i is not a source or the sink, total flow into i equals total flow out of a node. Mathematically,

$$\sum_{j \in I(i)} x_{ji} = \sum_{j \in O(i)} x_{ij} \quad \text{for all } i \in \mathcal{N} \setminus \{s, t\}.$$

Capacity constraints

$$0 \leq x_{ij} \leq c_{ij} \quad (i, j) \in \mathcal{A}.$$

General Maximum Flow Problems for single source and single sink (which arcs may carry factional units of flow)

$$\begin{aligned} \text{Maximise } & \sum_{i \in I(t)} x_{it} \\ \text{subject to } & \sum_{j \in I(i)} x_{ji} = \sum_{j \in O(i)} x_{ij} \quad \text{for all } i \in \mathcal{N} \setminus \{s, t\} \\ & 0 \leq x_{ij} \leq c_{ij} \quad (i, j) \in \mathcal{A}. \end{aligned}$$

Theorem. Consider a linear programming problem maximize$\{\mathbf{c}^T\mathbf{x} : A\mathbf{x} \leq b, \mathbf{x} \geq 0\}$ with an integer vector b such that the optimal value is finite. If the matrix A is a totally unimodular matrix in the sense that every square submatrix of A has determinant $+1, -1$ or 0, then

- All the extreme point of $\{A\mathbf{x} \leq b, \mathbf{x} \geq 0\}$ are integral vectors, and so, this linear programming problem has an integral optimal solution.
- Moreover, the integral solution of maximize$\{\mathbf{c}^T\mathbf{x} : A\mathbf{x} \leq b, \mathbf{x} \geq 0\}$ is also a solution of maximize$\{\mathbf{c}^T\mathbf{x} : A\mathbf{x} \leq b, \mathbf{x} \geq 0, \mathbf{x} \text{ is an integral vector}\}$.

Chapter 5

Conic Linear Programs

5.1 Convex Cones and Conic Programs

5.1.1 Inner Product Spaces

Let H be a finite dimensional space. We say a mapping $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{R}$ is an inner product in H if

(1) $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$;

(2) $\langle x, y \rangle = \langle y, x \rangle$;

(3) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$, for $\alpha \in \mathbb{R}$;

(4) $\langle x, x \rangle \geq 0$ for all $x \in H$, and $\langle x, x \rangle = 0$ if and only if $x = 0$.

A finite dimensional space H together with an inner product on it is called a finite dimensional inner product space. Examples:

- $H = \mathbb{R}^n$ where $\langle x, y \rangle = x^T y$ for all $x, y \in \mathbb{R}^n$;
- $H = \mathbb{R}^{p \times n}$ where $\langle X, Y \rangle = \mathrm{tr}\big(X^T Y\big)$ for all $X, Y \in \mathbb{R}^{p \times n}$;
- $H = S^n$ where $\langle X, Y \rangle = tr(XY)$ for all $X, Y \in S^n$.

Definition. (Cone) A set K in a finite dimensional inner product space is called a cone if, for all $x \in K$ and $\alpha \geq 0$, $\alpha x \in K$. In other words, K is a cone if, for any point $x \in K$, the ray

$$\mathbb{R}_+ x = \{\alpha x : \alpha \geq 0\}$$

is a subset of K.

Basic examples of cones:

1. Nonnegative cone: $K = \mathbb{R}^n_+$;
2. Second-order cone:

$$K = \mathrm{SOC}^n := \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : \sqrt{x_2^2 + \cdots + x_n^2} \leq x_1\};$$

3. Positive semi-definite cone:

$$K = \mathbb{S}^n_+ := \{X \in \mathbb{R}^{n\times n} : X \text{ is a symmetric positive semi-definite matrix}\};$$

4. A simple two dimensional cone:

$$K = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 x_2 \geq 0\}.$$

Definition. (Convex Cones) A convex cone is a cone that is also a convex set.

Theorem. (Alterntive Definition of a Convex Cone) A cone $K \subseteq \mathbb{R}^n$ is convex if and only if

$$x + y \in K \quad \text{for all } x, y \in K.$$

Proposition. The positive semi-definite cone S^n_+ is a convex cone.

Theorem. Let C_i be convex cones, $i = 1, \ldots, p$. Then, the following statements hold:

1. $\bigcap_{i=1}^p C_i$is a convex cone if $H_1 = H_2 = \cdots H_p = H$ and $C_i \subseteq H$.
2. $\prod_{i=1}^p C_i := C_1 \times C_2 \times \cdots C_p$ is also a convex cone.
3. The Minikowski sum

$$C_1 + C_2 + \cdots + C_p = \{x : x = \sum_{i=1}^p x_i, x_i \in C_i\}$$

is also a convex cone if $H_1 = H_2 = \cdots = H_p = H$.

Note: For two convex cones, $C_1 \cup C_2$ need not be convex!

Definition. (Dual Cone) Let K be a convex cone in an inner product space H. The (positive) dual cone is defined as K^* which is defined by

$$K^* = \{x \in H : \langle z, x\rangle \geq 0 \text{ for all } z \in K\}.$$

In the literature, there is also a notation called polar cone of K is denoted as K° and is defined by

$$K^\circ = -K^* = \{x \in H : \langle z, x\rangle \leq 0 \text{ for all } z \in K\}.$$

Proposition. The positive dual cone of positive semi-definite cones S^n_+ is itself (equivalently, the polar cone of positive semi-definite cone S^n_+ is $-S^n_+$).

Definition. We say a convex cone K is self-dual if $K^* = K$. The nonnegative cone, the second order cone and the positive semidefinite cone are all self-dual convex cones.

Let H be a finite dimensional inner produce space and let $K \subseteq H$ be a convex cone in H. A conic programming problem has the form

$$\begin{aligned} \text{Minimise}_{x\in H} \quad & \langle c, x\rangle \\ \text{subject to} \quad & \langle a_i, x\rangle = b_i, i = 1, \ldots, m, \\ & x \in K, \end{aligned}$$

where c and $a_i, i = 1, \ldots, m$ are elements of an inner product space $H, b_i, i = 1, \ldots, m$ are real numbers and K is a closed convex cone in H. The program is a convex program in which the only non-linearities appear in the cone constraint $x \in K$.

Some special cases of conic programming problems:

- For $H = \mathbb{R}^n, \langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v}$ and $K = \mathbb{R}^n_+$, then the associated conic programming problem is just the **linear programming problem**.

- For $H = \mathbb{R}^n, \langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^T \mathbf{v}$ and $K = \prod_{j=1}^p \mathrm{SOC}^{n_j}$ with $\sum_{j=1}^n n_j = n$, then the associated conic programming problem

$$\begin{aligned} \text{Minimise}_{x \in \mathbb{R}^n} \quad & \mathbf{c}^T \mathbf{x} \\ \text{subject to} \quad & \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, \\ & \mathbf{x} \in \prod_{j=1}^p \mathrm{SOC}^{n_j}, \end{aligned}$$

  is called a primal form (or conic form) of **second-order cone programming problem**. For $H = S^n, \langle A, B \rangle := \mathrm{tr}(AB)$ and $K = S^n_+$, then the associated conic programming problem

$$\begin{aligned} \text{Minimise}_{x \in S^n} \quad & \mathrm{tr}(CX) \\ \text{subject to} \quad & \mathrm{tr}(A_i X) = b_i, i = 1, \ldots, m, \\ & X \in S^n_+ \text{ (or } X \succeq 0), \end{aligned}$$

  is called a primal form (or conic form) of **semi-definite programming problem**.

5.2 Dual Problems of Conic Program and Duality Theory

Recall the form for a conic programming problem. Its dual problem reads

$$\begin{aligned} \text{Maximise}_{y \in \mathbb{R}^m} \quad & \sum_{i=1}^m y_i b_i \\ \text{subject to} \quad & c - \sum_{i=1}^m y_i a_i \in K^* \end{aligned}$$

The. pair rof problems (P) and (D) is called a primal and dual pair of conic problems. The dual problem of the previous special cases are as follows:

- **Linear Programming**:

$$\begin{aligned} \text{Maximise} \quad & \sum_{i=1}^m b_i y_i \\ \text{subject to} \quad & \mathbf{c} - \sum_{i=1}^m y_i \mathbf{a}_i \geq 0 \end{aligned}$$

- **Second-order Cone Programming**:

$$\begin{aligned} \text{Maximise}_{\mathbf{y}\in\mathbb{R}^m} \quad & \sum_{i=1}^m b_i y_i \\ \text{subject to} \quad & \mathbf{c} - \sum_{i=1}^m y_i \mathbf{a}_i \in \prod_{j=1}^p \mathrm{SOC}^{n_j} \end{aligned}$$

- **Semidefinite Programming**:

$$\begin{aligned} \text{Maximise}_{\mathbf{y}\in\mathbb{R}^m} \quad & \sum_{i=1}^m b_i y_i \\ \text{subject to} \quad & C - \sum_{i=1}^m y_i A_i \succ 0. \end{aligned}$$

Theorem. (Weak Duality) Let $x \in H$ be a feasible point to the problem (P) and let $\mathbf{y} \in \mathbb{R}^m$ be a feasible point to (D). Then, we have

$$\langle c, x \rangle \geq \sum_{i=1}^m b_i y_i.$$

In particular, we have $\min(P) \geq \max(D)$.

Definition. (Duality Gap) We define the duality gap ϑ between the problem (P) aand its dual problem (D) by

$$\vartheta := \min(P) - \max(D).$$

Theorem. (Strong Duality Theorem) Asume that both primal (P) and dual problems have nonempty feasible regions. Moreover, suppose that the primal problem (P) satisfies the following interior point condition: there exists $\overline{x} \in H$ such that

$$\langle a_i, \overline{x} \rangle = b_i, i = 1, \ldots, m, \text{ and } \overline{x} \in \mathrm{int}(K).$$

Then,

(1) $\min(P) = \max(D)$;

(2) the maximum in the dual problem (D) is attained.

5.3 Modelling with Conic Programs

In this section, we examine several important classes (second order cone programs, semi-definite programs, other conic programs) of conic programs and discuss specific modelling techniques for these classes.

5.3.1 Second-order Cone Programs

Note: A primal or a dual form of second order cone program is a convex program.

5.3.1.1 Secord-order Cone Representable Set

Case 1: Epigraph set of Euclidean norms. Note that

$$\|x\|_2 \iff \begin{bmatrix} t \\ \mathbf{x} \end{bmatrix} \in \mathrm{SOC}^{n+1}.$$

So,

$$\{(\mathbf{x}, t) : \|\mathbf{x}\|_2 \leq t\} = \left\{(\mathbf{x}, t) : \begin{bmatrix} t \\ \mathbf{x} \end{bmatrix} \in \mathrm{SOC}^{n+1}\right\}$$

Case 2: Sets described by norm inequalities. Consider a set Ω given by

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : \|A\mathbf{x} + \mathbf{b}\|_2 \leq \mathbf{c}^T\mathbf{x} + d\}$$

where A is an $(m \times n)$ matrix, $\mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ and $d \in \mathbb{R}$. Then, we see that

$$\Omega = \left\{\mathbf{x} \in \mathbb{R}^n : \begin{bmatrix} \mathbf{c}^T\mathbf{x} + d \\ A\mathbf{x} + \mathbf{b} \end{bmatrix} \in \mathrm{SOC}^{m+1}\right\}$$

Case 3: Epigraph set of square Euclidean norms The epigraph of the squared Euclidean norm can be described aas the intersection of a rotated quadratic cone with an affine hyperplane. Note that,

$$\begin{aligned}
&\|\mathbf{x}\|_2^2 \leq t = \frac{(t+1)^2 - (t-1)^2}{4} \\
\iff &\|\mathbf{x}\|_2^2 + \left(\frac{t-1}{2}\right)^2 \leq \left(\frac{t+1}{2}\right)^2, \quad t \geq 0 \\
\iff &\sqrt{\|\mathbf{x}\|_2^2 + \left(\frac{t-1}{2}\right)^2} \leq \frac{t+1}{2} \\
\iff &\left\|\begin{bmatrix} \mathbf{x} \\ \frac{t-1}{2} \end{bmatrix}\right\|_2 \leq \frac{t+1}{2} \iff \qquad \begin{bmatrix} \frac{t+1}{2} \\ \mathbf{x} \\ \frac{t-1}{2} \end{bmatrix} \in \mathrm{SOC}^{n+2}.
\end{aligned}$$

Case 4: Sets defined by convex quadratic inequality Assume $Q \in \mathbb{R}^{n \times n}$ is a symmetric positive semidefinite matrix. We consider a set Ω described by a convex inequality, that is,

$$\Omega = \{\mathbf{x} : \mathbf{x}^T Q\mathbf{x} + \mathbf{c}^T\mathbf{c} + r \leq 0\}.$$

The set Ω can be rewritten as

$$\mathbf{x}^T Q\mathbf{x} \leq -\mathbf{c}^T\mathbf{x} - r.$$

Since Q is symmetric positive semidefinite, there exists a matrix $F \in \mathbb{R}^{k \times n}$ such that

$$Q = F^T F.$$

Then

$$\mathbf{x}^T Q\mathbf{x} \leq -\mathbf{c}^T\mathbf{x} - r \iff \|F\mathbf{x}\|^2 \leq -\mathbf{c}^T\mathbf{x} - r.$$

So we have,

$$\begin{aligned}
&\{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}^T Q\mathbf{x} + \mathbf{c}^T\mathbf{x} + r \leq 0\} \\
&\qquad = \{\mathbf{x} \in \mathbb{R}^n : \|F\mathbf{x}\|_2^2 \leq -\mathbf{c}^T\mathbf{x} - r\} \\
&\qquad = \left\{\mathbf{x} \in \mathbb{R}^n : \begin{bmatrix} \frac{-\mathbf{c}^T\mathbf{x} - r + 1}{2} \\ F\mathbf{x} \\ \frac{-\mathbf{c}^T\mathbf{x} - r - 1}{2} \end{bmatrix} \in \mathrm{SOC}^{k+2}\right\}.
\end{aligned}$$

Case 5: Sets involving simple power functions Some sets described by power-like functions can be representable by second order cone, even though they need not be obvious at first glance. For example, we have

$$\left\{(x,t) \in \mathbb{R} \times \mathbb{R} : \frac{1}{x} \leq t, x > 0\right\} = \left\{(x,t) \in \mathbb{R} \times \mathbb{R} : \begin{bmatrix} x+t \\ x-t \\ 2 \end{bmatrix} \in \mathrm{SOC}^3\right\}.$$

5.3.1.2 Modelling with Second-Order Cone Programming

Consider a general convex quadratically constrained quadratic optimization problem (QCQP) can be written as

$$\begin{aligned} &\text{minimise } \mathbf{x}^T Q_0 \mathbf{x} + \mathbf{c}_0^T \mathbf{x} + r_0 \\ &\text{subject to } \mathbf{x}^T Q_i \mathbf{x} + \mathbf{c}_i^T \mathbf{x} + r_i \leq 0, \quad i = 1, \ldots, p, \end{aligned}$$

where all $Q_i \in \mathbb{R}^{n \times n}$ are symmetric positive semidefinite matrices. Let

$$Q_i = F_i^T F_i, \quad i = 0, \ldots, p,$$

where $F_i \in \mathbb{R}^{k_i \times n}$. We can equivalently rewrite a convex QCQP as a second-order cone program.

Special Case 1 of QCQP: Markowitz Protfolio Optimization AN important special case of the previous general model is the Markovitz portfolio optimization problem.

- One divides his/her capital over n possible investments (like stocks or bonds);
- x_i is the fraction of the investment capital one puts in investments i;
- r_i is the expected return on investment i;
- $\mathbf{x}^T \Sigma x$ is the risk (or volatility) associated with the portfolio $\mathbf{x}$, where Σ is a positive semi-definite matrix;
- $\mathbf{r}^T \mathbf{x}$ is the expected return on the portfolio.
- One wish to maximise the expected return given an upper bound γ on the tolerable risk.

The Markovitz portfolio optimization problem can be formulated as

$$\begin{aligned} \text{maximize} \quad & \mathbf{r}^T \mathbf{x} \\ \text{subject to} \quad & \mathbf{x}^T \Sigma \mathbf{x} \leq \gamma \\ & \mathbf{e}^T \mathbf{x} = 1 \\ & \mathbf{x} \geq 0, \end{aligned}$$

where $\mathbf{e} = (1, \ldots, 1)^T \in \mathbb{R}^n$. Let $\Sigma = GG^T$ (e.g., using a Cholesky or a eigenvalue decomposition). We then obtain a second-order cone programming formulation.

Special Case 2 of QCQP: Support Vector Machine Optimization formula for Support Vector Machine (SVM):

$$\begin{array}{rl} \text{Minimize}_{\ \mathbf{w}\in\mathbb{R}, \gamma\in\mathbb{R}} & \|\mathbf{w}\|_2^2 \\ \text{subject to} & \mathbf{w}^T\overline{\mathbf{x}_i} + \gamma \geq \quad 1, \quad \text{if } A \\ & \mathbf{w}^{\mathbf{T}}\overline{\mathbf{x}_i} + \gamma \leq -1, \quad \text{if } B \end{array}$$

5.3.2 Semi-definite Programs

The Trace Product

- $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for symmetric $(n \times n)$ matrices A, B;
- $\operatorname{tr}\big(A(xx^T)\big) = x^T A x$ for symmetric $(n \times n)$ matrices A;

Schur's Complement Let X be a symmetric $(p+q) \times (p+q)$ matrix given by

$$X = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}$$

where $A \in \mathbb{R}^{p\times p}$ is a symmetric matrix, $B \in \mathbb{R}^{p\times q}$ and $C \in \mathbb{R}^{q\times q}$ is a symmetric matrix.

If C is positive definite, then X is positive semi-definite if and only if

$$A - BC^{-1}B^T$$

is positive semi-definite.

5.3.2.1 Modelling with semi-definite programming

We first see that any linear programs and second-order cone programs can be regarded as a special semi-definite programs.

(1) Consider a linear program

$$\begin{array}{rl} \text{Minimize}_{\mathbf{x}\in\mathbb{R}^n} & \mathbf{c}^T\mathbf{x} \\ \text{Subject to} & \mathbf{a}_i^T\mathbf{x} = b_i, i = 1, \dots, m, \\ & \mathbf{x} \geq \mathbf{0}. \end{array}$$

Then, the above linear program can be written as

$$\begin{array}{rl} \text{Minimize}_{X\in S^n} & \displaystyle\sum_{j=1}^{n} c_j X_{jj} \\ \text{Subject to} & \displaystyle\sum_{j=1}^{n} a_{ij} X_{jj} = b_i, i = 1, \dots, m, \\ & X_{ij} = 0, i \neq j \\ & X = \begin{pmatrix} X_{11} & \cdots & X_{1n} \\ \vdots & \ddots & \vdots \\ X_{1n} & \cdots & X_{nn} \end{pmatrix} \succeq 0. \end{array}$$

(2) Consider a second order cone program

$$\begin{aligned} \text{Minimize}_{\mathbf{x}\in\mathbb{R}^n} \quad & \mathbf{c}^T\mathbf{x} \\ \text{Subject to} \quad & \mathbf{a}_i^T\mathbf{x} = b_i, i = 1, \dots, m, \\ & \mathbf{x} \in \mathrm{SOC}^n. \end{aligned}$$

Note that

$$\begin{aligned} x_1 \geq \sqrt{x_2^2 + \cdots + x_n^2} = \left\| \begin{pmatrix} x_2 \\ \vdots \\ x_n \end{pmatrix} \right\|_2 & \iff x_1^2 \geq \left\| \begin{pmatrix} x_2 \\ \vdots \\ x_n \end{pmatrix} \right\|, x_1 \geq 0, \\ & \iff \begin{pmatrix} x_1 & \mathbf{v}^T \\ \mathbf{v} & x_1 I_{n-1} \end{pmatrix} \succeq 0 \end{aligned}$$

where $\mathbf{v} = \begin{pmatrix} x_2 \\ \vdots \\ x_n \end{pmatrix}$. This occurs since in both cases $x_1 = 0$ (then all $x_2 = \cdots = x_n = 0$) and $x_1 \neq 0$ (by Schur's complement) the equivalence holds. So, second order cone program is a special semi-definite program with the form

$$\begin{aligned} \text{Minimize}_{X\in S^n} \quad & \sum_{j=1} c_j X_{1j} \\ \text{Subject to} \quad & \sum_{j=1}^{n} a_{ij} X_{1j} = b_i, i = 1, \dots, m \\ & X_{jj} = X_{11}, j = 1, \dots, n \\ & X_{jk=0}, j \neq k, 2 \leq j \leq n, 2 \leq k, \leq n \\ & X = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1n} \\ X_{12} & X_{22} & \cdots & X_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ X_{1n} & X_{2n} & \cdots & X_{nn} \end{pmatrix} \succeq 0. \end{aligned}$$

(3) Maximum eigenvalue minimization problem. Consider

$$\text{Minimize}_{\mathbf{x}\in\mathbb{R}^n} \lambda_{\max}\left(A_0 + \sum_{j=1}^{n} x_j A_j\right)$$

where A_0 and $A_j, j = 1, \dots, n$, are $(n \times n)$ symmetric matrices.

Note that $\lambda_m ax(X) \leq t$ if and only if

$$\max\{R_X(\mathbf{x}) : \mathbf{x} \neq 0\} \leq t$$

where $R_X(\mathbf{x})$ is the Rayleigh quotient of the matrix X defined as $R_X(\mathbf{x}) = \frac{\mathbf{x}^T X \mathbf{x}}{\|\mathbf{x}\|^2}$ with $\mathbf{x} \neq 0$. So, $\lambda_{\max}(X) \leq t$ is equivalent to, for all $\mathbf{x} \neq 0$

$$\mathbf{x}^T X \mathbf{x} \leq t\|\mathbf{x}\|_2^2,$$

which is further equivalent to

$$\begin{aligned}
&\mathbf{x}^T X \mathbf{x} \leq t\|\mathbf{x}\|_2^2 \text{ for all } \mathbf{x} \in \mathbb{R}^m \\
\iff &\mathbf{x}^T X \mathbf{x} \leq t\mathbf{x}^T I_m \\
\iff &\mathbf{x}^T (tI_m - X)\mathbf{x} \geq 0 \text{ for all } \mathbf{x} \in \mathbb{R}^m \\
\iff &tI_m - X \succeq 0.
\end{aligned}$$

Thus, the problem can be reformulated as a semidefinite programming problem:

$$\begin{aligned}
\text{Minimize}_{\mathbf{x}\in\mathbb{R}^n, t\in\mathbb{R}} \quad & t \\
\text{Subject to} \quad & -A_0 + \sum_{j=1}^n x_j(-A_j) + tI_m \succeq 0
\end{aligned}$$

(4) Consider the following fraction programs

$$\begin{aligned}
\text{Minimize}_{\mathbf{x}\in\mathbb{R}^n} \quad & \frac{(\mathbf{c}^T\mathbf{x} + r)^2}{\mathbf{d}^T\mathbf{x} + s} \\
\text{Subject to} \quad & \mathbf{a}^T\mathbf{x} = b_i, i = 1, \ldots, m.
\end{aligned}$$

Here, $\mathbf{c} \in \mathbb{R}^n, \mathbf{d} \in \mathbb{R}^n$ and $\mathbf{a} \in \mathbb{R}^n$, r, s, b_i are real numbers, and we assume that for all $\mathbf{x}$ with $\mathbf{a}^T\mathbf{x} = b_i, i = 1, \ldots, m$,

$$\mathbf{d}^T\mathbf{x} + s > 0.$$

Denote the feasible set

$$\Omega = \{\mathbf{x} : \mathbf{a}^T\mathbf{x} = b_i, i = 1, \ldots, m\}.$$

Note that (using Schur's complement)

$$\frac{(\mathbf{c}^T\mathbf{x} + r)^2}{\mathbf{d}^T\mathbf{x} + s} \leq t \text{ and } \mathbf{a}^T\mathbf{x} = b_i, i = 1, \ldots, m$$

if and only if

$$\begin{pmatrix} t & \mathbf{c}^T\mathbf{x} + r \\ \mathbf{c}^T\mathbf{x} + r & \mathbf{d}^T\mathbf{x} + s \end{pmatrix} \succeq 0 \text{ and } \mathbf{a}^T\mathbf{x} = b_i, i = 1, \ldots, m.$$

So, the fractional program can be reformulated as the following semi-definite programs:

$$\begin{aligned}
\text{Minimize}_{\mathbf{x}\in\mathbb{R}^n, t\in\mathbb{R}} \quad & t \\
\text{Subject to} \quad & \begin{pmatrix}
t & \mathbf{c}^T\mathbf{x} + r & 0 & \cdots & 0 \\
\mathbf{c}^T\mathbf{x} + r & \mathbf{d}^T\mathbf{x} + s & 0 & \cdots & \vdots \\
0 & 0 & b_1 - \mathbf{a}_1^T\mathbf{x} & \cdots & \vdots \\
\vdots & \vdots & 0 & \ddots & \vdots \\
0 & 0 & 0 & \cdots & b_m - \mathbf{a}_m^T\mathbf{x}
\end{pmatrix} \succeq 0
\end{aligned}$$

5.4 Integer Programs

A typical integer programming problem (IP) looks like

$$\begin{aligned}
\text{Minimize/Maximize} \quad & c_1x_1 + \cdots c_nx_n \\
\text{subject to} \quad & a_{11}x_1 + \cdots + a_{1n}x_n \leq b_1 \\
& \vdots \\
& a_{m1}x_1 + \cdots + a_{mn}x_n \leq n_m \\
x_i geq 0 \quad i = 1, \ldots, n & \\
x_i \text{ integers} \quad i \in I, &
\end{aligned}$$

where I is an nonempty subset of $\{1, \ldots, n\}$.

If $I \neq \{1, \ldots, n\}$, then we have a **mixed integer programming problem**. On the other hand, if $I = \{1, \ldots, n\}$ then we have a **pure integer programming problem**.
We present a few characterisations on integer programming problems.

- **Fixed Charge Problems.** The objective function is discontinuous.
- **Problems with either-Or Constraints.** In some situations, we are given two constraints of the form
$$f(x_1, \ldots, x_n) \leq 0 \text{ and } g(x_1, \ldots, x_n) \leq 0.$$
We want to ensure that *at least* one of the above two inequalities is satisfied, often call either-or constraints. In general, to handle these we choose a large number M so that
$$f(x_1, \ldots, x_n) \leq M \text{ and } g(x_1, \ldots, x_n) \leq M$$
holds for all values of $x_1, \ldots, x_n$ that satisfy the other constraints in the problem and introduce an auxiliary binary variable y. Then the above either-or constraints can be converted to
$$\begin{cases} f(x_1, \ldots, x_n) \leq My & \qquad y \in \{0, 1\} \\ g(x_1, \ldots, x_n) \leq M(1-y) & \end{cases}$$
- **K out of N Constraints Must Hold.** This is a generalisation of the previous situation where K constraints hold and $N - K$ do not. This can be solved similarly to the approach detailed above, but to all N constraints. We also need to introduce a summation constraint
$$\sum_{i=1}^{N} y_i = N - K.$$
The summation constraint forces $N-K$ of the y_i's to be 1 so that the corresponding $N-K$ constrains will have right hand sides $b_i + M$ and therefore not constrain the values of the x_i. The remaining K y_i values will be zero so that the right hand sides of those constraints will simply be b_i, the same as the original constraints.
- **Constraints with a Choice of Values.** The general formulation for such problem is
$$a_{11}x_1 + \cdots + a_{n1}x_n = b_1^{(1)} \text{ or } b_1^{(2)} \text{ or } \ldots \text{ or } b_1^{(k)}$$

and to reformulate this appropriately we introduce k binary variables y_i, one for each possible constraints value and replace the above constraint with

$$
\begin{gathered}
a_{11}x_1 + \cdots + a_{n1}x_n = \sum_{i=1}^{k} b_1^{(i)} y_i \\
\sum_{i=1}^{k} y_i = 1 \\
y_i \text{ binary} \qquad i = 1, 2 \ldots, k
\end{gathered}
$$

- **Binary Representations of General Integer Variables.** If all but a few variables in an integer programming problem are binary then it may be worthwhile to convert the remaining integer variables to binary ones as well and treat the problem with a method for binary integer programming problems. Suppose x is an integer variable with bounds

  $$0 \leq x \leq u, \quad \text{ where } 2^N \leq u < 2^{N+1}$$

  then we can replace x by its binary representation

  $$x = \sum_{i=0}^{N} 2^i y_i \quad \text{ with } y_i \in \{0, 1\}.$$

5.4.1 Branch and Bound Method

One approach to solving integer programming problems is to solve the corresponding linear programming problem, where we remove the conditions that the variables are integers (this is referred to as its *LP-relaxation*) and round the solution variables to the nearest feasible integer values. Of course if the solution to the LP-relaxation happens to be integers then the optimal solution to the original integer programming problem has been found.

The difficulty with this procedure is that none of the rounded solutions may be feasible, and the other difficulty is that it will probably not be optimal.

For now on, we will focus on integer programming problems with the form

$$
\begin{array}{rl}
\text{(IP) } \textit{Minimize} & c_1x_1 + \cdots + c_nx_n \\
\text{subject to} & a_{11}x_1 + \cdots + a_{1n}x_n \leq b_1 \\
& \vdots \\
& a_{m1}x_1 + \cdots + a_{mn}x_n \leq b_m \\
& x_i \geq 0, i = 1, \ldots, m \\
& x_i \text{ integers}, i = 1, \ldots, n,
\end{array}
$$

where all the coefficients $c_i, i = 1, \ldots, n$ (in the objective function) are all integers. If c_i are rational numbers, then we can multiply by a constant to force c_i to be integers.

5.4.1.1 Branch and Bound

At each step in the procedure we will divide the current LP-relaxation into 2 subproblems (this is called *branching*). These are solved (in our case by the Simplex method) and their optimal values give a bound as to the best possible solution that can be found for that subproblem. If there can be no integer solution in that subproblem which will be better than the current best integer solution then that branch is discarded (this is called *fathoming*).

Step 0. Solve the LP-relaxation of the problem. If the optimal solution is integer then the optimal solution to the original problem has also been found and the calculations are stopped. Otherwise treat this as the one remaining unfathomed subproblem and go to step 1.

Step 1. Branching: Among the remaining unfathomed subproblems, select one variable which takes non-integer value x, and branch from this subproblem by choosing one variable to be smaller than $\lfloor x \rfloor$ (the largest integer smaller than x) for one branch and greater than $\lceil x \rceil$ (the smallest integer larger than x) for the other branch.

Step 2. Bounding: For each new subproblem obtain its bound by applying the Simplex method to its LP-relaxation and rounding up the value of Z for the resulting optimal solution.

Step 3. Fathoming: For each new subproblem, apply the three fathoming test summarised below, and discard those subproblems that are fathomed by any of the tests.

Step 4. Optimality: If there are no remaining subproblems then the best integer solution found so far will be optimal. Otherwise return to step 1.

The Three Fathoming Tests. In the process of carrying out the branch and bound method suppose you have found an integer solution to one of the subproblems with value Z^* that is the best integer solution found so far (if no integer solution has yet been found let $Z^* = +\infty$).
A subproblem is fathomed (dismissed from further consideration) if

- Test 1: Its optimal value $> Z^*$ or
- Test 2: Its LP-relaxation has no feasible solutions or
- Test 3: The optimal solution for its LP-relaxation is integer.

The steps in this method are recorded on a tree, where the root of the tree is the inital LP relaxation, and the branching in Step 1 forms two new branches in the tree.

5.4.2 Cutting Plane Techniques

We now see one method that could potentially speed up the branch and bound method, called cutting plane techniques.

Cutting planes, sometimes called cuts or valid inequalities are linear inequalities that can be inferred from an integer or mixed integer constraint set. Cutting planes are added to the constraint set to "cut off" noninteger solutions of its LP relaxation, thus resulting in a tighter relaxation.

Consider an integer programming problem (IP) from before and it's relaxation (LP). One can solve (LP) via simplex method and obtain a solution $\mathbf{x}^*_{LP}$.

Definition. Consider an inequality with the form

$$\mathbf{w}^T\mathbf{x} + r \leq 0$$

where $\mathbf{w} \in \mathbb{R}^n$ and $r \in \mathbb{R}$. We say this inequality is a cutting plane for the integer programming problem (IP) if

- $\mathbf{w}^T\mathbf{x} + r \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$ which is feasible for (IP).
- $\mathbf{w}^T\mathbf{x}^*_{LP} + r > 0$ where $\mathbf{x}^*_{LP}$ is a solution for the linear programming relaxation problem.

From the definition, if a cutting plane is added to the linear program relaxation, then

- it will not cut off any feasible points for the integer program
- the solution obtained from solving the linear program relaxation will be cut off.

Thus, adding a cutting plane to the linear program relaxation will lead to tighter relaxation, that is, a solution with better lower bound.

Chapter 6

Solutions to Tutorial Problems

Note the problems are not available here. Please see course resources for reference.

6.1 Optimization Model Formulations

1. **Decision Variables**

   - x_1 - exterior paints to produce (in tons)
   - x_2 - interior paints to produce (in tons)

   Objective Function

   - $f(x_1, x_2) = 3000x_1 + 2000x_2$

   Constraints

   - $2x_1 + x_2 \leq 6$ (raw material A)
   - $x_2 + 2x_1 \leq 8$ (raw material B)
   - $x_2 \leq x_1 + 1$ (daily demand)
   - $x_2 \leq 2$ (maximum demand)
   - $x_1, x_2 \geq 0$ (hidden constraint)

   So the problem can be formulated as,

$$\begin{aligned} \max_{\mathbf{x}=(x_1,x_2)\in\mathbb{R}^2} \quad & 3000x_1 + 2000x_2 \\ \text{subject to} \quad & 2x_1 + x_2 \leq 6 \\ & x_2 + 2x_1 \leq 8 \\ & x_2 \leq x_1 + 1 \\ & x_2 \leq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

2. **Decision Variables**

   - x_1 - quantity of corn (in kgs)
   - x_2 - quantity of tankage (in kgs)
   - x_3 - quantity of alfalfa (in kgs)

   Objective Function

   - $f(x_1, x_2, x_3) = 42x_1 + 36x_2 + 30x_3$

   Constraints

   - $90x_1 + 20x_2 + 40x_3 \geq 200$ (carbohydrates)
   - $30x_1 + 80x_2 + 60x_3 \geq 180$ (proteins)
   - $10x_1 + 20x_3 + 60x_3 \geq 150$ (vitamins)
   - $x_1, x_2, x_3 \geq 0$ (hidden constraint)

So the problem can be formulated as,

$$\begin{aligned} \min_{\mathbf{x}=(x_1,x_2,x_3)\in\mathbb{R}^3} \quad & 42x_1 + 36x_2 + 30x_3 \\ \text{subject to} \quad & 90x_1 + 20x_2 + 40x_3 \geq 200 \\ & 30x_1 + 80x_2 + 60x_3 \geq 180 \\ & 10x_1 + 20x_3 + 60x_3 \geq 150 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

6.2 Mathematical Background

1. (a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. Observe that $\lambda_1 + \lambda_2 = \mathrm{tr}(A) = 1 + 4 > 0$ and $\lambda_1\lambda_2 = \det(A) = 4 - 4 = 0$. Hence either $\lambda_1 = 5, \lambda_2 = 0$ or $\lambda_1 = 0, \lambda_2 = 5$ so $\lambda_1, \lambda_2 \geq 0$. Thus A is positive semi-definite.

   (b) $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$. Observe that $\lambda_1\lambda_2 = \det(A) = 3 - 4 < 0$. Thus A is indefinite.

   (c) $A = \begin{pmatrix} 6 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 4 \end{pmatrix}$. Observe that $|6| > |1| + |2|, |3| > |1| + |1|, |4| > |2| + |1|$, so A is strictly diagonal dominant. Furthermore, it has positive diagonal entries. Thus A is positive definite.

   (d) $A = \begin{pmatrix} 6 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & -1 \end{pmatrix}$. Let $d_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $d_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Then, $d_1^T A d_1 = 6 > 0$ and $d_2^T A d_2 = -1$. Thus A is indefinite.

   (e) $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 6 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. The eigenvalues of A are $2, \lambda_1$ and λ_2. The eigenvalues λ_1 and λ_2 are eigenvalues of the submatrix $A_0 = \begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}$ which is positive definite as $\mathrm{tr}(A_0) = 7 > \det(A_0) = 2 > 0$. Thus A is positive definite.

   (f) $A = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$. Observe that A is diagonal dominant with non-negative diagonal elements. Thus A is positive semi-definite.

2. (a) SKIPPED

   (b) $\Omega = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$. Let $y, z \in \Omega$ and $\lambda \in [0, 1]$. Then

$$\begin{aligned} \|\lambda y + (1-\lambda)z\|_2 &\leq \|\lambda y\|_2 + \|(1-\lambda)z\|_2 \\ &= \lambda\|y\|_2 + (1-\lambda)\|z\|_2 \\ &< 1 \end{aligned}$$

   since $y, z \in \Omega \implies \|y\|, \|z\| < 1$ and $\lambda \in [0, 1]$. Hence Ω is convex.

(c) $\Omega = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2 \leq 1\} \cup \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0\}$. Consider $y = (-1, 0)$ and $z = (0, 10)$ then $y, z \in \Omega$. However the midpoint $(-\frac{1}{2}, 5) \notin \Omega$. Hence Ω is not convex.

3. SKIPPED

4. (1) SKIPPED

   (2) SKIPPED

   (3) $f(x_1, x_2, x_3) = x_1^4 + x_2^4 + x_3^4 - \cos(x_1^2 + x_2^2 + x_3^2)$. Observe that as $\|\mathbf{x}\| \to \infty$, $x_1^4 + x_2^4 + x_3^4 \to \infty$ and $\cos(x_1^2 + x_2^2 + x_3^2) \leq 1$ for all $\mathbf{x}$.So their difference approaches ∞. Hence f is coercive.

   (4) $f(x_1, x_2) = e^{x_1^2} + e^{x_2^2} - x_1^2 - x_2^2$. We have

$$\begin{aligned}
e^{x_1^2} + e^{x_2^2} - x_1^2 - x_2^2 &= e^{x_1^2} + e^{x_2^2} - (x_1^2 + x_1^2) \\
&= e^{x_1^2} + e^{x_2^2} - \|\mathbf{x}\|_2^2 \\
&\geq 2\left(e^{x_1^2} e^{x_2^2}\right)^{\frac{1}{2}} - \|\mathbf{x}\|_2^2 && \text{(using } a + b \geq 2\sqrt{ab}\text{)} \\
&= 2\left(e^{x_1^2 + x_2^2}\right)^{\frac{1}{2}} - \|\mathbf{x}\|_2^2 \\
&= 2e^{\frac{\|\mathbf{x}\|_2^2}{2}} - \|\mathbf{x}\|_2^2
\end{aligned}$$

   . Then since $\lim_{\|x\| \to \infty} \frac{e^{\frac{\|\mathbf{x}\|_2^2}{2}}}{\|\mathbf{x}\|_2^2} \to \infty$ then $f \to \infty$. Hence f is coercive.

5. SKIPPED

6. SKIPPED

7. (1) SKIPPED

   (2) $f(X) = \text{rank}(X)$ for all $X \in S^n$. Let $X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $Y = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Then $\text{rank}(X) = 1$ and $\text{rank}(Y) = 1$, but $\text{rank}(\frac{X+Y}{2}) = 2$. So $\text{rank}(\frac{X+Y}{2}) > \text{rank}(A) + \text{rank}(B)$. Thus f is not convex.

   (3) $f(X) = \det(X)$ for all $X \in S^n$. Similarly to above, $f(\frac{X+Y}{2}) = \frac{1}{4} > f(\frac{X}{2}) + f(\frac{Y}{2}) = 0 + 0$. Thus f is not convex.

6.3 Geometry of Linear Programs and Simplex Methods

1. (a) $\Omega_1 = \{\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1, x_i \geq 0, i = 1, 2, 3\}$. We see that $n = 3$ and $m = 1$, then hence an extreme point has $n - m = 2$ entries are zero. So candidates for extreme points are

$$\mathbf{x}_1^* = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{x}_2^* = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{x}_3^* = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}.$$

   Now $x_1 + x_2 + x_3 = 1 \iff \mathbf{a}_1^T \mathbf{x} = b_i$ with $\mathbf{a}_1 = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$ and $b_i = 1$. So $A = \mathbf{a}_1^T = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$. For $\mathbf{x}_1^*$, as only the first component is non-zero and $\{A_1\}$ is linearly independent then $\mathbf{x}_1^*$ is an extreme point of Ω_1. Similarly $\mathbf{x}_2^*$ and $\mathbf{x}_3^*$ are also extreme points of Ω_1.

   (b) $\Omega_2 = \{\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n : \sum_{i=1}^n x_i = 2, x_i \geq 0, i = 1, \ldots, n\}$. Similarly $m = 1$, so there are $n - 1$ zero entries in the extreme points. Then from above, it follows that each $2\mathbf{e}_i = \begin{pmatrix} 0 & \cdots & 2 & \cdots & 0 \end{pmatrix} \in \mathbb{R}^n$ is an extreme point of Ω_2.

(c) $\Omega_3 = \{\mathbf{x} = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + x_2 + x_3 = 1, x_2 + x_4 = 2, x_i \geq 0, i = 1, \ldots, 4\}$. Note that $n = 4$ and $m = 2$, so extreme points have $n - m = 2$ entries are zero. Then note that

$$\begin{aligned} x_1 + x_2 + x_3 = 1 &\iff \mathbf{a}_1^T \mathbf{x} = 1 \\ x_2 + x_4 = 2 &\iff \mathbf{a}_2^T \mathbf{x} = 1 \end{aligned}$$

where $\mathbf{a}_1 = \begin{pmatrix} 1 & 1 & 1 & 0 \end{pmatrix}^T$ and $\mathbf{a}_2 = \begin{pmatrix} 0 & 1 & 0 & 1 \end{pmatrix}^T$. So, $A = \begin{pmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$.

i. If $x_1 = x_2 = 0$, then $x_3 = 1, x_4 = 2$. As $\{A_3, A_4\}$ are linearly independent, $\begin{pmatrix} 0 & 0 & 1 & 2 \end{pmatrix}$ is an extreme point.

ii. If $x_1 = x_3 = 0$, then $x_2 = 1, x_4 = 1$. As $\{A_2, A_4\}$ are linearly independent, $\begin{pmatrix} 0 & 1 & 0 & 1 \end{pmatrix}$ is an extreme point.

iii. If $x_1 = x_4 = 0$, then $x_2 = 2$ and $x_3 = -1$ which is not in the feasible region.

iv. If $x_2 = x_3 = 0$, then $x_1 = 1, x_4 = 2$. As $\{A_1, A_4\}$ are linearly independent, $\begin{pmatrix} 1 & 0 & 0 & 2 \end{pmatrix}$ is an extreme point.

v. If $x_2 = x_4 = 0$, then $x_2 + x_4 \neq 2$ which is impossible.

vi. If $x_3 = x_4 = 0$, then $x_2 = 2, x_1 = -1$ which is not in the feasible region.

Thus extreme points for Ω_3 are $\mathbf{x} = \begin{pmatrix} 0 & 0 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 2 \end{pmatrix}$.

2. SKIPPED

3. SKIPPED

4. **Decision Variables**

- x_1 - number of inkjet printers
- x_2 - number of laser printers

Objective Function

- $f(x_1, x_2) = 30x_1 + 25x_2$

Constraints

- $x_1 + 2x_2 \leq 40$ (total hours)
- $3x_1 + x_2 \leq 45$ (storage space)
- $x_2 \leq 12$ (market capacity)
- $x_1, x_2 \geq 0$ (hidden constraint)

So the problem can be formulated as,

$$\begin{aligned} \max_{\mathbf{x}=(x_1,x_2)\in\mathbb{R}^2} \quad & 30x_1 + 25x_2 \\ \text{subject to} \quad & x_1 + 2x_2 \leq 40 \\ & 3x_1 + x_2 \leq 45 \\ & x_2 \leq 12 \\ & x_1, x_2 \geq 0. \end{aligned}$$

In terms of standard form, we have

$$\begin{aligned} \min_{\mathbf{x}=(x_1,x_2)\in\mathbb{R}^2} \quad & -30x_1 - 25x_2 \\ \text{subject to} \quad & x_1 + 2x_2 + s_1 = 40 \\ & 3x_1 + x_2 + s_2 = 45 \\ & x_2 + s_3 = 12 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0. \end{aligned}$$

To use simplex method, we first rewrite the above standard form as

$$\begin{aligned} \text{Minimise} \quad & z \\ \text{subject to} \quad & z + 30x_1 + 25x_2 = 0 \\ & x_1 + 2x_2 + s_1 = 40 \\ & 3x_1 + x_2 + s_2 = 45 \\ & x_2 + s_3 = 12 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0. \end{aligned}$$

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	
	1	30	25	0	0	0	0	
s_1	0	1	2	1	0	0	40	non-basic variables:
s_2	0	3	1	0	1	0	45	x_1, x_2
s_3	0	0	1	0	0	1	12	

With our entering variable x_1 and outgoing variable s_2:

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	
	1	0	15	0	-10	0	-450	
s_1	0	0	$\frac{5}{3}$	1	$-\frac{1}{3}$	0	25	non-basic variables:
x_1	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	15	s_2, x_2
s_3	0	0	1	0	0	1	12	

Now entering variable x_2 and outgoing variable s_3:

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	
	1	0	0	0	-10	-15	-630	
s_1	0	0	0	1	$-\frac{1}{3}$	$-\frac{5}{3}$	5	non-basic variables:
x_1	0	1	0	0	$\frac{1}{3}$	$-\frac{1}{3}$	11	s_2, s_3
x_2	0	0	1	0	0	1	12	

Thus, an optimal solution is $x_1^* = 11, x_2^* = 12$.

6.4 Duality and Sensitivity Analysis for Linear Programs

1. **Primal Problem**

$$\begin{aligned} \text{minimize} \quad & -5x_1 + 3x_2 \\ \text{subject to} \quad & x_1 - 2x_2 + x_3 = 1 \\ & 3x_1 + 4x_2 + x_4 = 12 \\ & 7x_1 - x_2 + x_5 = 14 \\ & x_i \geq 0, i = 1, \ldots, 5 \end{aligned}$$

Dual Problem

$$\begin{aligned} \text{maximize} \quad & y_1 + 12y_2 + 14y_3 \\ \text{subject to} \quad & y_1 + 3y_2 + 7y_3 \leq -5 \\ & -2y_1 + 4y_2 - y_3 \leq 3 \\ & y_1, y_2, y_3 \leq 0 \end{aligned}$$

2. (a) **Primal Problem**

$$\begin{aligned} \text{minimize} \quad & 2x_1 - x_2 + x_3 \\ \text{subject to} \quad & 5x_1 - 4x_2 + 3x_2 \leq 2 \\ & 5x_1 - x_3 \geq -4 \\ & x_1 + x_2 = 1 \\ & x_1 + 2x_2 + x_3 \leq 14 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Dual Problem

$$\begin{aligned} \text{maximize} \quad & 2y_1 - 4y_2 + y_3 + 14y_4 \\ \text{subject to} \quad & 2 - 5y_1 - 5y_2 - y_3 - y_4 \geq 0 \\ & -1 + 4y_1 - y_3 - 2y_4 \geq 0 \\ & 1 - 3y_1 + y_2 - y_4 \geq 0 \\ & y_1, y_4 \leq 0, y_2 \geq 0, y_3 \text{ unrestricted} \end{aligned}$$

(b) **Primal Problem** **Dual Problem**

$$\begin{aligned} \text{minimize} \quad & -5x_1 - 2x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 1 \\ & 2x_1 - x_2 \geq 2 \\ & x_1 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{maximize} \quad & y_1 + 2y_2 \\ \text{subject to} \quad & 5 - y_1 - 2y_2 \geq 0 \\ & -2 - y_1 + y_2 = 0 \\ & y_1 \leq 0, y_2 \geq 0 \end{aligned}$$

3. i) **Primal Problem** **Dual Problem**

$$\begin{aligned} (EP) \text{ minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 = 4 \\ & 4x_1 + 2x_2 + x_4 = 12 \\ & -x_1 + x_2 + x_5 = 1 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

$$\begin{aligned} (ED) \text{ maximize} \quad & 4y_1 + 12y_2 + y_3 \\ \text{subject to} \quad & -1 - y_1 - 4y_2 + y_3 \geq 0 \\ & -1 - 2y_1 - 2y_2 - y_3 \geq 0 \\ & y_1, y_2, y_3 \leq 0. \end{aligned}$$

ii) By complementary slackness, we have the following equations then substituting $\mathbf{x}^*$ and multiplying,

$$\left.\begin{aligned} x_1^*(-1 - y_1^* - 4y_2^* + y_3^*) = 0 \\ x_2^*(-1 - 2y_1^* - 2y_2^* - y_3^*) = 0 \\ x_3^*(-y_1^*) = 0 \\ x_4^*(-y_2^*) = 0 \\ x_5^*(-y_3^*) = 0 \end{aligned}\right\} \implies \left.\begin{aligned} -1 - y_1^* - 4y_2^* + y_3^* = 0 \\ -1 - 2y_1^* - 2y_2^* - y_3^* = 0 \\ -y_3^* = 0 \end{aligned}\right\} \implies \left\{\begin{aligned} y_1^* &= -\frac{1}{3} \\ y_2^* &= -\frac{1}{6} \\ y_3^* &= 0 \end{aligned}\right\}$$

So, the solution for the dual problem is $\mathbf{y}^* = (-\frac{1}{3}, -\frac{1}{6}, 0)$.

iii) Optimal value for $(EP) = -x_1^* - x_2^* = -\frac{10}{3}$
Optimal value for $(ED) = 4y_1^* + 12y_2^* + y_3^* = (-\frac{4}{3}) + 12(-\frac{1}{6}) = -\frac{10}{3}$.

4. Consider the perturbed problem

$$\begin{aligned} \text{Minimize} \quad & -3x_1 - 5x_2 \\ \text{subject to} \quad & x_1 + s_1 = 4 + r_1 \\ & 2x_2 + s_2 = 12 + r_2 \\ & 3x_1 + 2x_2 + s_3 = 18 + r_3 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0 \end{aligned}$$

Then computing the simplex table reads,

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	r_1	r_2	r_3	
	1	3	5	0	0	0	0	0	0	0	
s_1	0	1	0	1	0	0	4	1	0	0	non-basic variables:
s_2	0	0	2	0	1	0	12	0	1	0	x_1, x_2
s_3	0	3	2	0	0	1	18	0	0	1	

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	r_1	r_2	r_3	
	1	3	0	0	$-\frac{5}{2}$	0	-30	0	$-\frac{-5}{2}$	0	
s_1	0	1	0	1	0	0	4	1	0	0	non-basic variables:
x_2	0	0	1	0	$\frac{1}{2}$	0	6	0	$\frac{1}{2}$	0	x_1, s_2
s_3	0	3	0	0	-1	1	6	0	-1	1	

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	r_1	r_2	r_3	
	1	0	0	0	$-\frac{3}{2}$	-1	-36	0	$-\frac{3}{2}$	-1	
s_1	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2	1	$\frac{1}{3}$	$-\frac{1}{3}$	non-basic variables:
x_2	0	0	1	0	$\frac{1}{2}$	0	6	0	$\frac{1}{2}$	0	s_3, s_2
x_1	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2	0	$-\frac{1}{3}$	$\frac{1}{3}$	

So we have

$$\begin{aligned}
\text{optimal value } &= -36 - \frac{3}{2}r_2 - r_3 \\
s_1 &= 2 + r_1 + \frac{1}{3}r_2 - \frac{1}{3}r_3 \\
x_2 &= 6 + \frac{1}{2}r_2 \\
x_1 &= 2 - \frac{1}{3}r_2 + \frac{1}{3}r_3
\end{aligned}$$

Provided that each of the above equations are ≥ 0, then we note $r_1 = 3.8 - 4 = -0.2, r_2 = 12.2 - 12 = 0.2$ and $r_3 = 17.9 - 17 = -0.1$. The equations are satisfied. So we have

$$\begin{aligned}
\text{optimal value } &= -36 - \frac{3}{2}(0.2) - (-0.1) = -36.2 \\
x_1 &= 2 - \frac{1}{3}(0.2) + \frac{1}{3}(-0.1) = 1.9 \\
x_2 &= 6 + \frac{1}{2}(0.2) = 6.1
\end{aligned}$$

So expected change in objective value is -0.2 and the new solution is $(1.9, 6.1)$.

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6.5 Interior Point Method for Linear Programs and Applications for Linear Programs

1. (a) **Primal Problem**

$$\begin{aligned}
(P_0) \quad \underset{\mathbf{x}=(x_1,x_2,x_3,x_4)\in\mathbb{R}^4}{\text{Minimize}} \quad & -2x_1 - 3x_2 \\
\text{subject to} \quad & x_1 + x_3 = 1 \\
& x_2 + x_4 = 1 \\
& x_1, x_2, x_3, x_4 \geq 0.
\end{aligned}$$

Dual Problem

$$\begin{aligned}
(D_0) \quad \underset{\mathbf{x}=(x_1,x_2,x_3,x_4)\in\mathbb{R}^4}{\text{Maximize}} \quad & y_1 + y_2 \\
\text{subject to} \quad & -2 - y_1 \geq 0 \\
& -3 - y_2 \geq 0 \\
& -y_1, -y_2 \geq 0.
\end{aligned}$$

(b) The μ-pertubed optimality condition is

$$(*) \begin{cases}
x_1 + x_3 = 1 & w_1 = -2 - y_1 \\
x_2 + x_4 = 1 & w_2 = -3 - y_2 \\
x_1, x_2, x_3, x_4 \geq 0 & w_3 = -y_1 \\
x_1 w_1 = \mu & w_4 = -y_2 \\
x_2 w_2 = \mu & w_1, w_2, w_3, w_4 \geq 0 \\
x_3 w_3 = \mu & \\
x4 w_4 = \mu &
\end{cases}$$

(c) For any $\mathbf{x}(\mu), \mu > 0$, on the central path, it satisfies the μ-perturbed optimality system (*) for some $\mathbf{y} \in \mathbb{R}^2$ and $\mathbf{w} \in \mathbb{R}^4$. We now solve (*) to obtain $\mathbf{x}(\mu)$.

(1) As $x_1 w_1 = \mu$ and $w_1 = -2 - y_1$, we have $x_1(-2 - y_1) = \mu$. Moreover, $x_3 w_4 = \mu$ and $w_3 = -y_1$, so $x_3(-y_1) = \mu$. Note that $x_1 + x_3 = 1$ so $(1 - x_1)(-y_1) = \mu$ so $y_1 = \frac{\mu}{-(1-x_1)}$. Substituting gives $x_1(-2 + \frac{\mu}{1-x_1}) = \mu$. Multiplying both sides gives,

$$\begin{aligned}
x_1[-2(1-x_1) + \mu] &= \mu(1 - x_1) \\
2x_1^2 + (\mu - 2)x_1 &= \mu - \mu x_1 \\
2x_1^2 + (2\mu - 2)x_1 &= \mu \\
x_1^2 + (\mu - 1)x_1 &= \frac{\mu}{2} \\
\left[x_1 + \frac{\mu - 1}{2}\right]^2 &= \frac{\mu}{2} + \frac{(\mu-1)^2}{4} \\
x_1 &= -\frac{\mu - 1}{2} \pm \sqrt{\frac{\mu}{2} + \frac{(\mu-1)^2}{4}}
\end{aligned}$$

As $x_1 \geq 0$, we have

$$x_1 = -\frac{\mu - 1}{2} + \sqrt{\frac{\mu}{2} + \frac{(\mu-1)^2}{4}}$$

$$\begin{aligned}
x_3 = 1 - x_1 &= 1 + \frac{\mu - 1}{2} - \sqrt{\frac{\mu}{2} + \frac{(\mu-1)^2}{4}} \\
&= \frac{\mu + 1}{2} - \sqrt{\frac{\mu}{2} + \frac{(\mu-1)^2}{4}}.
\end{aligned}$$

(2) Similarly, as $x_2 w_2 = \mu$ and $w_2 = -3 - y_2$ we get

$$x_2 = -\frac{2\mu - 3}{6} + \sqrt{\frac{\mu}{3} + \frac{(2\mu-3)^2}{36}} \text{ and } x_4 = 1 - x_2 = \frac{2\mu + 3}{6} - \sqrt{\frac{\mu}{3} + \frac{(2\mu-3)^2}{36}}$$

Thus $\mathbf{x}(\mu) = (x_1(\mu), x_2(\mu), x_3(\mu), x_4(\mu))$ as functions as above.

(d) Let $\mu \to 0$, then $x_1(\mu) \to 1, x_2(\mu) \to 1, x_3(\mu) \to 0, x_4(\mu) \to 0$. So $\mathbf{x}(\mu) = \mathbf{x}^* = (1, 1, 0, 0)$. We now see that $\mathbf{x}^*$ is an optional solution of (P_0). Clearly, it is a feasible point. Next, take any feasible point $\mathbf{x}$ of (P_0) then let $\mathbf{x} = (x_1, x_2, x_3, x_4)$. Then $x_1 \leq 1$ and $x_2 \leq 1$ and so

$$-2x_1 - 3x_2 \geq (-2) \cdot 1 + (-3) \cdot 1 = -5 = (-2)x_1^* + (-3)x_2^*.$$

So $\mathbf{x}^*$ is an optimal solution of (P_0).

2. Note that $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$ with $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$. So we can write

$$\begin{aligned}
\underset{\mathbf{x} \in \mathbb{R}^n, t_i \in \mathbb{R}, i=1,\ldots,n}{\text{Maximize}} \quad & \sum_{i=1}^n t_i \\
\text{subject to} \quad & |x_i| \leq t_i, \quad i = 1, \ldots, n \\
& A\mathbf{x} = \mathbf{b}.
\end{aligned}$$

Now notice that $|x_i| \leq t_i \iff \max\{x_i, -x_i\} \leq t_i \iff x_i \leq t_i$ and $-x_i \leq t_i$. So the problem can be reformulated as the following linear program

$$\begin{aligned} \underset{\mathbf{x}\in\mathbb{R}^n, t_i\in\mathbb{R}, i=1,\dots,n}{\text{Maximize}} \quad & \sum_{i=1}^{n} t_i \\ \text{subject to} \quad & x_i - t_i \le 0, \quad i = 1, \dots, n \\ & -x_i - t_i \le 0, \quad i = 1, \dots, n \\ & A\mathbf{x} = \mathbf{b}. \end{aligned}$$

3. Let x_{ij} be the number of kilowatt-hours (kwh) of electricity sent from power plant i to city j for $i = \{1, 2, 3\}$ and $j = \{1, 2, 3, 4\}$. Then the linear programming problem is

$$\begin{aligned} \text{Minimize} \quad & 4x_{11} + 3x_{12} + 10x_{13} + 9x_{14} \\ & 9x_{21} + 6x_{22} + 13x_{23} + 7x_{24} \\ & 12x_{31} + 9x_{32} + 16x_{33} + 6x_{34} \\ \text{subject to} \quad & \sum_{j=1}^{4} x_{1j} \le 35; \quad \sum_{i=1}^{3} x_{i1} \ge 45; \\ & \sum_{j=1}^{4} x_{2j} \le 50; \quad \sum_{i=1}^{3} x_{i2} \ge 20; \\ & \sum_{j=1}^{4} x_{3j} \le 40; \quad \sum_{i=1}^{3} x_{i3} \ge 30; \\ & \qquad\qquad\qquad \sum_{i=1}^{3} x_{i4} \ge 30; \\ & x_{ij} \ge 0 \text{ for } i = 1, 2, 3 \text{ and } j = 1, 2, 3, 4. \end{aligned}$$

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6.6 Network Problems

1. We apply Dijkstra's algorithm to find the shortest path between nodes S and T. Let P be the permanent label set in each step of Dijkstra's algorithm. Then we have

(1)

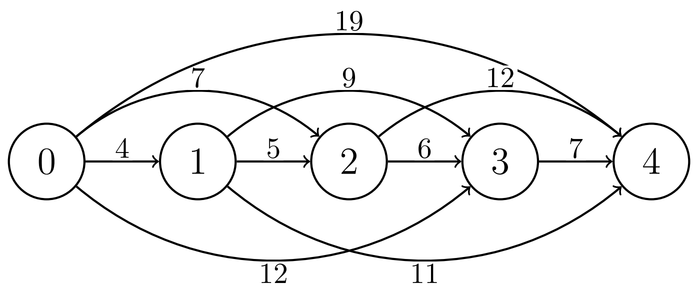
	P	distance to S
Step 0	$P = \{S\}$	0
Step 1	$P = \{S, A\}$	0, 8
Step 2	$P = \{S, A, B\}$	0, 8, 10
Step 3	$P = \{S, A, B, C\}$	0, 8, 10, 14
Step 4	$P = \{S, A, B, C, D\}$	0, 8, 10, 14, 16
Step 5	$P = \{S, A, B, C, D, E\}$	0, 8, 10, 14, 16, 17
Step 6	$P = \{S, A, B, C, D, E, F\}$	0, 8, 10, 14, 16, 17, 21
Step 7	$P = \{S, A, B, C, D, E, F, G\}$	0, 8, 10, 14, 16, 17, 21, 22
Step 8	$P = \{S, A, B, C, D, E, F, G, T\}$	0, 8, 10, 14, 16, 17, 21, 22, 27

So the shortest distance between S and T is 27 and the shortest path is $S \to A \to D \to G \to T$.

(2) SKIPPED

2. An arc (i, j) from node i to node j corresponds to purchasing a tractor at the end of year i and trading it at the end of year j, and the length of the arc (i, j) is the net discounted cost associated with this strategy. Running Dijkstra gives a shortest path of $0 \to 1 \to 3$ with cost 29.

3. Here an arc (i, j) from node i to node j corresponds to purchasing a car at the end of the year i and sell it at the end of year j, and the length of the arc (i, j) is the net discounted cost associate with this strategy.



Running Dijkstra's algorithm results in the shortest path of $0 \to 1 \to 4$ with distance 15.

4. Note that finding the most reliable route from S to T is equivalent to finding the shortest path from node S to node T in the network where weights are set to $-\ln(w_i)$. Then the shortest path from node S to node T is $S \to A \to C \to F \to T$ which is also the most reliable route.

5. The linear programming problem formulation is

$$\begin{array}{rl}
\text{Maximize} & x_{SA} + x_{SB} + x_{SC} \\
\text{subject to} & x_{SA} = x_{AD} + x_{AF} \\
& x_{SB} + x_{CB} + x_{DB} = x_{BC} + x_{BD} + x_{BE} \\
& x_{SC} + x_{BC} = x_{CB} + x_{CE} \\
& x_{AD} + x_{BD} = x_{DB} + x_{DF} + x_{DG} \\
& x_{BC} + x_{CE} = x_{EG} \\
& x_{AF} + x_{DF} + x_{GF} = x_{FG} + x_{FT} \\
& x_{DG} + x_{EG} + x_{FG} = x_{GF} + x_{GT} \\
& \begin{array}{ll}
0 \le x_{SA} \le 6 & 0 \le x_{DB} \le 3 \\
0 \le x_{SB} \le 5 & 0 \le x_{DG} \le 3 \\
0 \le x_{SC} \le 3 & 0 \le x_{DF} \le 4 \\
0 \le x_{AD} \le 3 & 0 \le x_{EG} \le 4 \\
0 \le x_{AF} \le 6 & 0 \le x_{FT} \le 8 \\
0 \le x_{BC} \le 4 & 0 \le x_{GF} \le 4 \\
0 \le x_{BD} \le 5 & 0 \le x_{FG} \le 2 \\
0 \le x_{BE} \le 4 & 0 \le x_{CE} \le 5
\end{array}
\end{array}$$

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6.7 Conic Programs

6.7.1 Introduction, Geometry, Duality Theory

1. Let $C = \{(\mathbf{x}, t) : \|\mathbf{x}\| \leq t\}$. Take any $(\mathbf{x}, t) \in C$ and let $\alpha \geq 0$. Then $\alpha(\mathbf{x}, t) \in C$ because $\|\alpha\mathbf{x}\| = |\alpha| \cdot \|\mathbf{x}\| = \alpha\|\mathbf{x}\| \leq at$. So, C is a cone. Moreover, let $(\mathbf{x}_1, t_1) \in C, (\mathbf{x}_2, t_2) \in C$ and $\lambda \in [0, 1]$. Then

$$\begin{aligned}
\|\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2\| &\leq \|\lambda\mathbf{x}_1\| + \|(1-\lambda)\mathbf{x}_2\| && \text{(triangle inequality)}\\
&= |\lambda| \cdot \|\mathbf{x}_1\| + |1-\lambda| \cdot \|\mathbf{x}_2\| && \text{(homogenous propery for norm)}\\
&= \lambda \cdot \|\mathbf{x}_1\| + (1-\lambda)\|\mathbf{x}_2\| && (\lambda \in [0,1])\\
&\leq \lambda \cdot t_1 + (1-\lambda)t_2 && (\text{as } (\mathbf{x}_1, t_1), (\mathbf{x}_2, t_2) \in C)
\end{aligned}$$

So $\lambda(\mathbf{x}_1, t_1) + (1-\lambda)(\mathbf{x}_2, t_2) \in C$. So C is convex and hence C is a convex cone.

2. Let $\Omega = \{X \in S^n_+ : |X_{ij}| \leq 1, \text{ for all } i, j = 1, \dots, n\}$.

   - Closed. Let $X_k \in \Omega$ and $X_k \to X$. Then $X_k \in S^n_+$ and $|(X_k)_{ij}| \leq 1$. In particular for all $\mathbf{d} \in \mathbb{R}^n$, $\mathbf{d}^T X_k \mathbf{d} \geq 0$ and $|(X_k)_{ij}| \leq 1$. Let $k \to +\infty$, then $\mathbf{d}^T X \mathbf{d} \geq 0$ for all $\mathbf{d} \in \mathbb{R}^n$ and $|X_{ij}| \leq 1$. So $X \in \Omega$, so Ω is closed.
   - Convex. Let $\Omega_0 = \{X \in \mathbb{R}^{n\times n} : |X_{ij}| \leq 1, \text{ for all } i, j = 1, \dots, n\}$. Then $\Omega = S^n_+ \cap \Omega_0$. Let $X_1, X_2 \in \Omega_0$ and $\lambda \in [0, 1]$. Then for all $i, j = 1, \dots, n$

     $$|(X_1)_{ij}| \leq 1 \text{ and } |(X_2)_{ij}| \leq 1.$$

     Then

     $$\begin{aligned}
     |\lambda(X_1)_{ij} + (1-\lambda)(X_2)_{ij}| &\leq \lambda|(X_1)_{ij}| + (1-\lambda)|(X_2)_{ij}|\\
     &\leq \lambda 1 + (1-\lambda)\\
     &= 1
     \end{aligned}$$

     So, Ω_0 is convex. Since S^n_+ is convex, then the intersection of two convex sets is convex, i.e. Ω is convex.
   - Bounded. For all $X \in \Omega$, $|X_{ij} \leq 1, i, j = 1, \dots, n$. So, $\|X\|_F = \left[\sum_{i,j} |X_{ij}|^2\right]^{\frac{1}{2}} \leq n$ for all $X \in \Omega$. Thus $\Omega \subseteq \{X : \|X\|_F \leq n\}$ and so Ω is a bounded set.

3. SKIPPED

4. (1) Let C_1, C_2 be convex cones such that $C_1 \subseteq C_2$. Let $x \in C_2^*$ then $\langle x, c_2\rangle \geq 0$ for all $c_2 \in C_2$. As $C_1 \subseteq C_2$ this implies $\langle x, c_1\rangle \geq 0$ for all $c_1 \in C_1$. So $x \in C_1^*$. Thus $C_2^* \subseteq C_1^*$.

   (2) We show forward inclusion. Let $x \in (C_1 + C_2)^*$ then for all $c_1 \in C_1, c_2 \in C_2$, $\langle x, c_1 + c_2\rangle \geq 0$. Letting $c_1 = 0 \in C_1$, then $\langle x, c_2\rangle \geq 0$ for all $c_2 \in C_2$, so $x \in C_2^*$. Similarly for $x \in C_1^*$. Thus $x \in C_1^* \cap C_2^*$.

   For reverse, since $x \in C_1^* \cap C_2^*$ then $\langle x, c_1\rangle \geq 0$ and $\langle x, c_2\rangle \geq 0$. Summing these equations gives the result.

5. (1) The interior point condition means that the system has a solution (feasible region is non-empty). This can be satisfied with $X = \begin{pmatrix} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0 \end{pmatrix}$.

(2) The objective function can be rewritten as $2X_{12}$, since we take the trace of CX where X is a symmetric matrix. For any feasible point X, we have

$$X = \begin{pmatrix} 1 & X_{12} & X_{13} \\ X_{12} & 1 & X_{23} \\ X_{13} & X_{23} & 1 \end{pmatrix} \succeq 0$$
$$\begin{pmatrix} 1 & X_{12} \\ X_{12} & 1 \end{pmatrix} \succeq 0$$

and so $X_{12}^2 \leq 1$ so $X_{12} \geq -1$. So its objective value is $2X_{12} \geq -2$, so optimal value of $(P) \geq -2$.

On the other hand, let $X = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, then X is feasible and $\operatorname{tr}(CX) = -2$. So optimal value ≤ -2, thus optimal value must be $= 2$.

(3) **Primal Problem**

$$\begin{aligned} (P) \text{ minimize} \quad & \operatorname{tr}(CX) \\ \text{subject to} \quad & \operatorname{tr}(A_iX) = 1, i = 1, 2, 3 \\ & X \succeq 0 \end{aligned}$$

where A_i is the zero matrix except $A_{ii} = 1$.

Dual Problem

$$\begin{aligned} (D) \text{ maximize} \quad & y_1 + y_2 + y_3 \\ \text{subject to} \quad & \begin{pmatrix} -y_1 & 1 & 0 \\ 1 & -y_2 & 0 \\ 0 & 0 & -y_3 \end{pmatrix} \succeq 0. \end{aligned}$$

(4) The optimal value of $(D) = -2$ as the primal problem and dual have non-empty feasible regions, and the interior point condition is satisfied. By the strong duality theorem for conic programs, this implies that the optimal value of (D) is the optimal value of (P).

6. (1) Let $C_1 = \mathrm{SOC}^3 = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq \sqrt{x_2^2 + x_3^2}\}$ and $C_2 = \mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$. Then $K = C_1 \times C_2$. Now,

$$\begin{aligned} K^* &= (C_1 \times C_2)^* \\ &= C_1^* \times C_2^* \qquad\qquad = C_1 \times C_2 \\ &= K. \end{aligned}$$

as $C_1^* = (\mathrm{SOC}^3)^* = \mathrm{SOC}^3 = C_1$ and $C_2^* = (\mathbb{R}_+)^* = \mathbb{R}_+ = C_2$. So K is a self-dual cone.

(2)

$$\begin{aligned} \text{maximize} \quad & 1 \cdot y_1 + 0 \cdot y_2 \\ \text{subject to} \quad & \begin{pmatrix} -y_2 \\ -1 - y_1 \\ -y_2 \\ -y_1 \end{pmatrix} \in K^* = K. \end{aligned}$$

(3) For the primal problem, take any $\mathbf{x}$ which is feasible. Then $x_1 + x_3 = 0$ and $x_1 \geq \sqrt{x_2^2 + x_3^2}$ so $x_1^2 \geq x_2^2 + x_3^2 = x_2^2 + x_1^2 \implies x_2 = 0$. So the optimal value of the primal problem is 0.

Now take any feasible point for the dual. Since $\begin{pmatrix} -y_2 \\ -1-y_1 \\ -y_2 \\ -y_1 \end{pmatrix} \in \mathrm{SOC}^3$ we have

$$\begin{aligned} -y_2 &\geq \sqrt{(-1-y_1)^2 + (-y_2)^2} \\ (-y_2)^2 &\geq (-1-y_1)^2 + (-y)^2 \\ y_1 &= -1. \end{aligned}$$

So optimal value is -1 since for any feasible point $y_1 = -1$. Thus duality gap is optimal value of primal - optimal value of dual gives $0 - (-1) = 1$.

(4) We note that the primal problem does not satisfy the interior point condition which causes the duality gap to be non-zero. Indeed the interior point condition requires that the system has a solution. However if $\mathbf{x} \in \operatorname{int} K$, then $x_1 > \sqrt{x_2^2 + x_3^2} \geq |x_3|$ which contradicts with $x_1 + x_3 = 0$.

6.7.2 Modelling, Numerical Methods and Applications

1. (a) **Primal Problem**

$$\begin{aligned} (P_0)\ \text{Minimize}_{X \in S^2} \quad & \operatorname{tr}(CX) \\ \text{Subject to} \quad & X_{ii} = 1, i = 1, 2, \\ & X \succeq 0 \end{aligned}$$

where $C = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

Dual Problem

$$\begin{aligned} (D_0)\ \text{Maximise} \quad & y_1 + y_2 \\ \text{subject to} \quad & c - y_1 A_1 - y_2 A_2 \\ & = \begin{pmatrix} 1-y_1 & 2 \\ 2 & 1-y_2 \end{pmatrix} \succeq 0. \end{aligned}$$

where $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

(b) The logarithmic barrier problem (P_μ) for (P_0) is, for each $\mu > 0$,

$$\begin{aligned} (P_\mu)\ \text{Minimise}_{X \in S^2} \quad & \operatorname{tr}(CX) + \mu\phi_{\mathrm{SDP}}(X) \\ \text{subject to} \quad & X_{ii} = 1, i = 1, 2 \end{aligned}$$

$$\text{where} \quad \phi_{\mathrm{SDP}}(X) = \begin{cases} -\log\det(X) & \text{if } X \succeq 0 \\ +\infty & \text{else.} \end{cases}$$

(c) Any minimiser $X(\mu)$ of (P_μ) satisfies the optimality condition

$$\begin{aligned} \operatorname{tr}(A_i X) &= 1, i = 1, 2 \\ X &\succeq 0 \\ c - \sum_{i=1}^{2} y_i A_i &= \mu X^{-1} \end{aligned}$$

for some $\mathbf{y} = (y_1, y_2) \in \mathbb{R}^2$. We now solve the condition to obtain $X(\mu)$. From the last equation,

$$c - \sum y_i A_i)X = \mu I_2$$

$$\begin{pmatrix} 1-y_1 & 2 \\ 2 & 1-y_2 \end{pmatrix} \begin{pmatrix} X_{11} & X_{12} \\ X_{12} & X_{22} \end{pmatrix} = \begin{pmatrix} \mu & 0 \\ 0 & \mu \end{pmatrix}$$

This then shows that

$$(1-y_1)+2X_{12}=(1-y_1)X_{11}+2X_{12}=\mu$$
$$(1-y_1)X_{12}+2=(1-y_1)X_{12}+2X_{22}=0$$

Rearranging the first gives

$$(1-y_1)=\frac{-2}{X_{12}}.$$

Substituting this into the other equation gives,

$$\begin{aligned}\frac{-2}{X_{12}}+2X_{12}&=\mu && \text{(since } X_{11}=X_{22}=1)\\ -2+2X_{12}^2&=\mu X_{12}\\ X_{12}^2-\frac{\mu}{2}X_{12}&=1\\ (X_{12}-\frac{\mu}{4})^2&=1+\frac{\mu^2}{16}\\ X_{12}&=\frac{\mu}{4}\pm\sqrt{1+\frac{\mu^2}{16}}\end{aligned}$$

Then $X=\begin{pmatrix}1 & X_{12}\\ X_{12} & 1\end{pmatrix}\succ 0 \implies \det X>0 \implies X_{12}<1$. Thus

$$X(\mu)=\begin{pmatrix}1 & \frac{\mu}{4}-\sqrt{1+\frac{\mu^2}{16}}\\ \frac{\mu}{4}-\sqrt{1+\frac{\mu^2}{16}} & 1\end{pmatrix} \quad \text{with } \mu>0.$$

(d) When $\mu\to 0$, $X_{12}\to -1$ so $X^*=\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}$. Note that X^* is feasible since $X^*_{ii}=1, i=1,2$ and $X^*\succeq 0$. Moreover, for all X which is feasible for (P_0) we have $X_{12}^2\leq 1 \implies -1\leq X_{12}\leq 1$ (since $\det X\geq 0$). Thus,

$$\begin{aligned}\mathrm{tr}(CX)&=2+4X_{12}\\ &\geq 2+4(-1) && -2=\mathrm{tr}(CX^*).\end{aligned}$$

So, for all feasible X for (P_0), $\mathrm{tr}(CX)\geq tr(CX^*)$. This shows that X^* is an optimal solution for (P_0).

2. Observe that the problem is equivalent to

(1)

$$\begin{aligned}\text{Minimise} \quad & t\\ \text{subject to} \quad & x_1^2+x_2^2+x_3^2-x_1-t\leq 0\\ & x_1^2+x_2^2+x_3^2\leq 1 && x_i\geq 0, i=1,2,3.\end{aligned}$$

Also note that, $x_1^2+x_2^2+x_3^2=\|\mathbf{x}\|_2^2$. So we again reformulate this as

$$\begin{aligned}\text{Minimise} \quad & t\\ \text{subject to} \quad & \|\mathbf{x}\|_2^2\leq x_1+t \iff \|\mathbf{x}\|^2\leq\left(\frac{x_1+t+1}{2}\right)^2-\left(\frac{x_1+t-1}{2}\right)^2 \iff \left\|\begin{bmatrix}\mathbf{x}\\ \frac{x_1+t-1}{2}\end{bmatrix}\right\|_2\leq\frac{x_1+t}{2}\\ & \|\mathbf{x}\|_2^2\leq 1 \iff \|\mathbf{x}\|_2\leq 1\\ & \mathbf{x}\geq 0.\end{aligned}$$

which is a conic program (indeed a second order cone program).

$$\begin{aligned}
\text{Minimise} \quad & t \\
\text{subject to} \quad & \begin{bmatrix} \frac{(x_1+t+1)}{2} \\ \mathbf{x} \\ \frac{(x_1+t-1)}{2} \end{bmatrix} \in \mathrm{SOC}^5 \\
& \frac{1}{\mathbf{x}} \in \mathrm{SOC}^4
\end{aligned}$$

$$x_i \in \mathrm{SOC}^1, i = 1, 2, 3.$$

(2) This can be rewritten as

$$\text{Minimise}_{\mathbf{x}\in\mathbb{R}^3} \left\| \begin{bmatrix} x_1 - 1 \\ x_2 - 2 \\ x_3 - 1 \end{bmatrix} \right\|_2 + 2 \left\| \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \\ x_3 - 2 \end{bmatrix} \right\|$$

Which is then equivalent to

$$\begin{aligned}
\text{Minimise} \quad & t_1 + 2t_2 \\
\text{subject to} \quad & \left\| \begin{bmatrix} x_1 - 1 \\ x_2 - 2 \\ x_3 - 1 \end{bmatrix} \right\|_2 \le t_1 \\
& \left\| \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \\ x_3 - 2 \end{bmatrix} \right\|_2 \le t_2
\end{aligned}$$

which is a conic program (indeed a second order cone program)

$$\begin{aligned}
\text{Minimise} \quad & t_1 + 2t_2 \\
\text{subject to} \quad & \left\| \begin{bmatrix} t_1 \\ x_1 - 1 \\ x_2 - 2 \\ x_3 - 1 \end{bmatrix} \right\|_2 \in \mathrm{SOC}^4 \\
& \left\| \begin{bmatrix} t_2 \\ x_1 - 1 \\ x_2 - 1 \\ x_3 - 2 \end{bmatrix} \right\|_2 \in \mathrm{SOC}^4
\end{aligned}$$

3. (1) SKIPPED

   (2) SKIPPED

   (3) The problem is equivalent to the following semi-definite program

$$\begin{aligned}
\text{Minimise} \quad & x_1 \\
\text{subject to} \quad & \begin{pmatrix} x_1 & 1 \\ 1 & x_2 \end{pmatrix} \succeq 0.
\end{aligned}$$

4. SKIPPED

5. Using the fact that, for any symmetric matrix $A \in S^n$ and $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a}^T A\mathbf{a} = \mathrm{tr}\big(A\mathbf{a}\mathbf{a}^T\big)$, we have

$$\mathbf{x}^T C_0 \mathbf{x} = \mathrm{tr}\big(C_0 \mathbf{x}\mathbf{x}^T\big)$$
$$x_i^2 = \mathbf{x}^T E_i \mathbf{x} = \mathrm{tr}\big(E_i \mathbf{x}\mathbf{x}^T\big)$$

where E_i is the diagonal matrix whose (i,i)-th element is one and zero otherwise. Therefore, (QP) can be rewritten as

$$\begin{aligned} \text{Minimise}_{X \in S^n} \quad & \mathrm{tr}(C_0 X) \\ \text{subject to} \quad & \mathrm{tr}(E_i X) = 1, i = 1, \ldots, n \\ & = X \in \Omega = \{X \in S^n : X = \mathbf{x}\mathbf{x}^T \text{ for some } \mathbf{x} \in \mathbb{R}\}. \end{aligned}$$

Further, note that $\Omega \subseteq S_+^n$ (the cone of all positive semi-definite matrices). Thus, we see that $f_1^* = \min(QP) \geq \min(AP) = f_2^*$, since (AP)'s first constraint can be written as $\mathrm{tr}(E_i X) = 1, i = 1, \ldots, n$. Finally the conclusion follows by noticing $\mathrm{tr}(E_i X) = X_{ii}, i = 1, \ldots, n$ by the construction of E_i.

6.8 Integer Programs

1. Let x_i be the number of works employed on line $i, i = 1, 2$. Then the problem can be formulated as

$$\begin{aligned} \text{Minimise} \quad & 1000y_1 + 500x_1 + 2000y_2 + 900x_2 \\ \text{subject to} \quad & 20x_1 + 50x_2 \geq 120 \\ & 30x_1 + 35x_2 \geq 150 \\ & 40x_1 + 45x_2 \geq 200 \\ & x_1 \leq 7y_1 \\ & x_2 \leq 7y_2 \\ & x_1, x_2 \text{ are nonnegative integers} \qquad y_1, y_2 \in \{0, 1\} \end{aligned}$$

2. SKIPPED

3. SKIPPED

4. SKIPPED

5. (1) From the final simplex table, as the entries associated with the non-basic variables are all non-positive in the cost row, the solution can be found by letting the nonbasic variables $x_3 = x_4 = 0$. Then the solution is $x_1^* = \frac{25}{14}, x_2^* = \frac{23}{7}, x_3^* = x_4^* = 0$.

   (2) $\mathbf{x}^*$ is not a solution as x_1^*, x_2^* are not integers.

   (3) The two constraints of the final simplex table read as follows:

$$x_2 + \left(\frac{2}{7}\right) x_3 + \left(-\frac{1}{7}\right) x_4 = \frac{23}{7}$$
$$x_1 + \left(\frac{-3}{14}\right) x_3 + \left(\frac{5}{14}\right) x_4 = \frac{25}{14}$$

Further simplifying gives the following expressions.

$$(x_2 - x_4) + \left(\frac{2}{7}x_3 + \frac{6}{7}\right)x_4 = 3 + \frac{2}{7} \implies \frac{2}{7}x_3 + \frac{6}{7}x_4 - \frac{2}{7} = 3 - (x_2 - x_4)$$
$$(x_1 - x_3) + \left(\frac{11}{14}x_3 + \frac{5}{14}x_4\right) = 1 + \frac{11}{14} \implies \frac{11}{14}x_3 + \frac{5}{14}x_4 - \frac{11}{14} = 1 - (x_1 - x_3)$$

This shows that for any point feasible for the integer program, one must have

$$x_2 - x_4 = x_2 + 0 \cdot x_3 - x_4 \leq 3$$
$$x_1 - x_3 = x_1 - x_3 + 0 \cdot x_4 \leq 1$$

These are the Gomory cuts generated by the first and second constraint of the linear program relaxation.